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TEST 1

READING PASSAGE 1



Why We Need To Protect Polar Bears

Polar bears are being increasingly **threatened** by the

effects of **climate** change, but their disappearance could have **far-reaching consequences**. They are **uniquely adapted** to the **extreme** conditions of the Arctic Circle, where **temperatures** can **reach** -40°C . One reason for this is that they have up to 11 centimetres of fat **underneath** their skin. Humans with **comparative** levels of **adipose tissue** would be considered **obese** and would be likely to **suffer from diabetes** and heart **disease**. Yet the polar bear experiences no such consequences.

A 2014 study by Sin Ping Liu and **colleagues** **sheds light on** this **mystery**. They compared the **genetic** structure of polar

polar= close to or relating to the North Pole or the South Pole
threaten= to be likely to harm or destroy something
climate= the typical weather conditions in a particular area
far-reaching= having a great influence or effect
consequence= result, effect, outcome
uniquely= in a way that is different from anything or anyone else
adapt= to gradually change your behaviour and attitudes in order to be successful in a new **situation**
extreme= very unusual and severe or serious
temperature= a measure of how hot or cold a place or thing is
reach= if something reaches a particular rate, amount etc, it increases until it is at that rate or amount
underneath= under, beneath, below
comparative= relative, proportional
adipose= relating to animal fat
tissue= the material forming animal or plant cells
obese= very fat in a way that is unhealthy
suffer from something= to have a particular disease or medical condition, especially for a long time
diabetes= a serious disease in which there is too much sugar in your blood
disease= an illness which affects a person, animal, or plant
colleague= coworker, partner, teammate, associate
shed light on something= to make something easier to understand, by providing new or better information
mystery= an event, situation etc that people do not understand or cannot explain because they do not know enough about it
genetic= relating to genes or genetics

bears with that of their closest **relatives** from a warmer climate, the brown bears. This allowed them to **determine** the **genes** that have allowed polar bears to **survive** in one of the **toughest** environments on Earth. Liu and his colleagues found the polar bears had a gene known as APoB, which reduces levels of **low-density lipoproteins** (LDLs) — a form of 'bad' **cholesterol**. In humans, **mutations** of this gene are **associated with** increased risk of heart disease. Polar bears may therefore be an important study model to understand heart disease in humans.

The **genome** of the polar bear may also provide the **solution** for another condition, one that particularly affects our older generation: **osteoporosis**. This is a disease where bones show reduced density, usually caused by **insufficient** exercise, reduced **calcium intake** or food **starvation**. Bone tissue is **constantly** being **remodelled**, meaning that bone is added or removed, depending on **nutrient availability** and the stress that the bone is under. Female polar bears, however, **undergo** extreme conditions during every **pregnancy**. Once autumn comes around, these females will dig **maternity dens** in the snow and will remain there throughout the winter, both before and after the birth of their **cubs**. This process results in about six months of fasting, where the female bears have to keep themselves and their cubs alive, **depleting** their own calcium and calorie **reserves**. Despite this, their bones remain strong and **dense**.

Physiologists Alanda Lennox and Allen Goodship found an **explanation** for this **paradox** in 2008. They discovered that pregnant bears were able to increase the density of their bones before they started to build their dens. In addition, six months later, when they finally **emerged** from the den with their cubs, there was no evidence of **significant** loss of bone density. **Hibernating** brown bears do not have this **capacity** and must therefore **resort to** major bone **reformation** in the

relative= a member of your family = relation
determine= decide, conclude, establish, finalize
survive= to continue to live after an accident, war, or illness
gene= a part of a cell in a living thing that controls what it looks like, how it grows, and how it develops. People get their genes from their parents
tough= hard, dangerous, threatening, harsh
density= the degree to which an area is filled with people or things
lipoproteins= any of a group of soluble proteins that combine with and transport fat or other lipids in the blood plasma
cholesterol= a chemical substance found in your blood
mutation= change, alteration, transformation, modification
be associated with somebody or something= to be related to a particular subject, activity etc

genome= all the genes in one type of living thing
solution= answer, key, explanation
osteoporosis= a medical condition in which your bones become weak and break easily
insufficient= not enough, lacking, inadequate, deficient
calcium= a silver-white metal that helps to form teeth, bones, and chalk
intake= the amount of food, drink etc that you take into your body
starvation= hunger, food shortage, famine
constantly= continuously, frequently, repetitively
remodel= to change the shape, structure, or appearance of something, especially a building
nutrient= a chemical or food that provides what is needed for plants or animals to live and grow
availability= the state of being able to be used, bought, or found
undergo= experience, feel, suffer, go through
pregnancy= when a woman has a baby growing inside her body
maternity= relating to a woman who is pregnant or who has just had a baby
den= the home of some animals, for example lions or foxes
cub= the baby of a wild animal such as a lion or a bear
deplete= to reduce the amount of something that is present or available
reserve= a supply of something kept to be used if it is needed
dense= thick, solid, compressed, condensed

physiologist= a person who studies physiology
explanation= reason, account, clarification
paradox= a situation that seems strange because it involves two ideas or qualities that are very different
emerge= to appear or come out from somewhere
significant= large, considerable, major, big
hibernate= if an animal hibernates, it sleeps for the whole winter
capacity= ability, capability, power
resort to something= to do something bad, extreme, or difficult because you cannot think of any other way to deal with a problem
reformation= when something is completely changed in order to improve it

following spring. If the **mechanism** of bone remodelling in polar bears can be understood, many **bedridden** humans, and even astronauts, could potentially benefit.

The medical benefits of the polar bear for humanity certainly have their importance in our **conservation** efforts, but these should not be the only factors taken into consideration. We tend to want to protect animals we think are intelligent and **possess** emotions, such as elephants and primates. Bears, on the other hand, seem to be **perceived** as stupid and in many cases violent. And yet **anecdotal** evidence from the field challenges those **assumptions**, suggesting for example that polar bears have good problem-solving abilities. A male bear called GoGo in Tennoji Zoo, Osaka, has even been **observed** making use of a tool to **manipulate** his environment. The bear used a tree branch on **multiple** occasions to **dislodge** a piece of meat hung out of his reach. Problem-solving ability has also been witnessed in wild polar bears, although not as obviously as with GoGo. A calculated move by a male bear involved running and jumping onto **barrels** in an attempt to get to a photographer standing on a **platform** four metres high.

In other studies, such as one by Alison Ames in 2008, polar bears showed **deliberate** and focussed manipulation. For example, Ames observed bears putting objects in piles and then knocking them over in what appeared to be a game. The study demonstrates that bears are capable of **agile** and **thought-out** behaviours. These examples suggest bears have greater **creativity** and problem-solving abilities than previously thought.

As for emotions, while the evidence is once again anecdotal, many bears have been seen to hit out at ice and snow — seemingly out of **frustration** — when they have just missed out on a kill. Moreover, polar bears can form **unusual** relationships with other species, including playing with the dogs used to pull **sleds** in the Arctic. **Remarkably**, one hand-raised polar bear called Agee has formed a close relationship with her owner Mark Dumas to the point where they even swim together. This is even more **astonishing** since polar bears are known to **actively hunt** humans in the wild.

If climate change were to lead to their **extinction**, this would mean not only the loss of **potential breakthroughs** in human medicine, but more importantly, the disappearance of an intelligent, **majestic** animal.

mechanism= a system or a way of behaving that helps a living thing to avoid or protect itself from something difficult or dangerous
bedridden= unable to leave your bed, especially because you are old or ill

conservation= the protection of natural things such as animals, plants, forests etc, to prevent them from being spoiled or destroyed
possess= have, own, hold, keep
perceive= see, understand, identify, recognize
anecdotal= consisting of short stories based on someone's personal experience
assumption= something that you think is true although you have no definite proof
observe= see, witness, detect, spot
manipulate= to make someone think and behave exactly as you want them to, by skilfully deceiving or influencing them
multiple= many, numerous, various
dislodge= to force or knock something out of its position
barrel= a large curved container with a flat top and bottom, made of wood or metal, and used for storing beer, wine etc
platform= a tall structure built so that people can stand or work above the surrounding area

deliberate= purposeful, conscious, intentional, calculated, planned
agile= able to move quickly and easily
thought-out= planned and organized carefully, well etc
creativity= imagination, originality, inventiveness

frustration= the feeling of being annoyed, upset, or impatient, because you cannot control or change a situation, or achieve something
unusual= strange, odd. Bizarre
sled= a small vehicle used for sliding over snow, often used by children or in some sports
remarkably= amazingly, outstandingly, extraordinarily, surprisingly
astonishing= amazing, surprising, shocking
actively= in a way that involves doing a lot of practical things
hunt= to chase animals and birds in order to kill or catch them
in the wild= in natural and free conditions, not kept or controlled by people

extinction= when a particular type of animal or plant stops existing
potential= possible, latent, probable, likely
breakthrough= an important new discovery in something you are studying, especially one made after trying for a long time
majestic= very big, impressive, or beautiful

TEST 1

READING PASSAGE 2



The Step Pyramid Of Djoser

The **pyramids** are the most famous **monuments** of **ancient** Egypt and still hold **enormous interest** for people in the present day. These **grand, impressive tributes** to the memory of the Egyptian kings have become linked with the country even though other cultures, such as the Chinese and Mayan, also built pyramids. The **evolution** of the pyramid form has been written and argued about for centuries. However, there is no question that, **as far as Egypt is concerned**, it began with one monument to one king designed by one **brilliant architect**: the Step Pyramid of Djoser at Saqqara.

pyramid= a large stone building with four triangular (=three-sided) walls that slope in to a point at the top, especially in Egypt and Central America
monument= a building, statue, or other large structure that is built to remind people of an important event or famous person
ancient= early, antique, olden
enormous= huge, vast, giant
interest= attraction, fascination, appeal
grand= outstanding, impressive, majestic
impressive= something that is impressive makes you admire it because it is very good, large, important etc
tribute= something that you say, do, or give in order to express your respect or admiration for someone
evolution= development, growth, progression, advancement
as far as something is concerned: about something, with regard to something
brilliant= excellent, great, wonderful
architect= someone whose job is to design buildings

Djoser was the first king of the Third Dynasty of Egypt and the first to build in stone. **Prior to** Djoser's **reign**, **tombs** were **rectangular** monuments made of dried **clay brick**, which covered **underground passages** where the **deceased** person was **buried**. For reasons which remain unclear, Djoser's main official, whose name was Imhotep, **conceived** of building a taller, more impressive tomb for his king by **stacking** stone **slabs** on top of one another, **progressively** making them smaller, to form the shape now known as the Step Pyramid. Djoser is thought to have reigned for 19 years, but some **historians** and **scholars attribute** a much longer time for his rule, owing to the number and size of the monuments he built.

The Step Pyramid has been **thoroughly examined** and investigated over the last century, and it is now known that the building process went through many different stages. Historian Marc Van de Mieroop comments on this, writing 'Much **experimentation** was involved, which is especially clear in the **construction** of the pyramid in the center of the **complex**. It had several plans ... before it became the first Step Pyramid in history, piling six levels on top of one another ... The weight of the enormous mass was a challenge for the builders, who placed the stones at an **inward incline** in order to prevent the monument breaking up.'

When finally completed, the Step Pyramid rose 62 meters high and was the tallest structure of its time. The complex in which it was built was the size of a city in ancient Egypt and included a temple, **courtyards**, **shrines**, and living **quarters** for the **priests**. It covered a region of 16 hectares and was surrounded by a wall 10.5 meters high. The wall had 13 false doors cut into it with only one true **entrance** cut into the south-east corner; the entire wall was then ringed by a **trench** 750 meters long and 40 meters wide. The **false** doors and the trench were **incorporated** into the complex to **discourage** unwanted visitors. If someone wished to enter, he or she would have needed to know in advance how to find the location of the true opening in the wall. Djoser was so proud of his **accomplishment** that he broke the tradition of having only his own name on the monument and had Imhotep's name **carved** on it as well.

The burial **chamber** of the tomb, where the king's body was laid to rest, was dug **beneath** the **base** of the pyramid,

prior to= before
reign= the period when someone is king, queen, or emperor
tomb= a stone structure above or below the ground where a dead person is buried
rectangular= having the shape of a rectangle
clay= a type of heavy sticky earth that can be used for making pots, bricks etc
brick= a hard block of baked clay used for building walls, houses etc
underground= below the surface of the earth
passage= way, road, channel, route, path
deceased= dead
bury= to put someone who has died in a grave
conceive= think of, consider, perceive
stack= load, pile, heap
slab= a thick flat piece of a hard material such as stone
progressively= increasingly, gradually
historian= someone who studies history, or the history of a particular thing
scholar= an intelligent and well-educated person
attribute= assign, attach, ascribe

thoroughly= completely, totally
examine= investigate, check, analyze, explore
experimentation= the process of testing various ideas, methods etc to find out how good or effective they are
construction= building, creation
complex= a group of buildings, or a large building with many parts, used for a particular purpose
inward= inner, interior, hidden
incline= a slope

courtyard= an open space that is completely or partly surrounded by buildings
shrine= a place that is connected with a holy event or holy person, and that people visit to pray
quarter= an area of a town
priest= someone who is specially trained to perform religious duties and ceremonies in the Christian church
entrance= a door, gate etc that you go through to enter a place
trench= a long narrow hole dug into the surface of the ground
false= untrue, incorrect, wrong
incorporate= to include something as part of a group, system, plan etc
discourage= to persuade someone not to do something, especially by making it seem difficult or bad
accomplishment= something successful or impressive that is achieved after a lot of effort and hard work
carve= to cut a pattern or letter on the surface of something

chamber= hall, boardroom, meeting room
beneath= under, underneath, below
base= the lowest part or surface of something

surrounded by a vast **maze** of long **tunnels** that had rooms off them to discourage **robbers**. One of the most **mysterious discoveries** found inside the pyramid was a large number of stone **vessels**. Over 40,000 of these vessels, of various forms and shapes, were discovered in storerooms off the pyramid's underground passages. They are **inscribed** with the names of rulers from the First and Second Dynasties of Egypt and made from different kinds of stone. There is no agreement among scholars and **archaeologists** on why the vessels were placed in the tomb of Djoser or what they were supposed to **represent**. The archaeologist Jean-Philippe Lauer, who **excavated** most of the pyramid and complex, believes they were originally stored and then given a 'proper burial' by Djoser in his pyramid to **honor** his **predecessors**. There are other historians, however, who claim the vessels were dumped into the **shafts** as yet another **attempt** to prevent grave robbers from getting to the king's burial chamber.

Unfortunately, all of the **precautions** and **intricate** design of the underground **network** did not prevent ancient robbers from finding a way in. Djoser's grave goods, and even his body, were stolen at some point in the past and all archaeologists found were a small number of his **valuables overlooked** by the **thieves**. There was enough left throughout the pyramid and its complex, however, to **astonish** and amaze the archaeologists who excavated it.

Egyptologist Miroslav Verner writes, 'Few monuments hold a place in human history as significant as that of the Step Pyramid in Saqqara ... It can be said without **exaggeration** that this pyramid complex **constitutes** a **milestone** in the evolution of monumental stone architecture in Egypt and in the world as a whole.' The Step Pyramid was a **revolutionary** advance in architecture and became the **archetype** which all the other great pyramid builders of Egypt would follow.

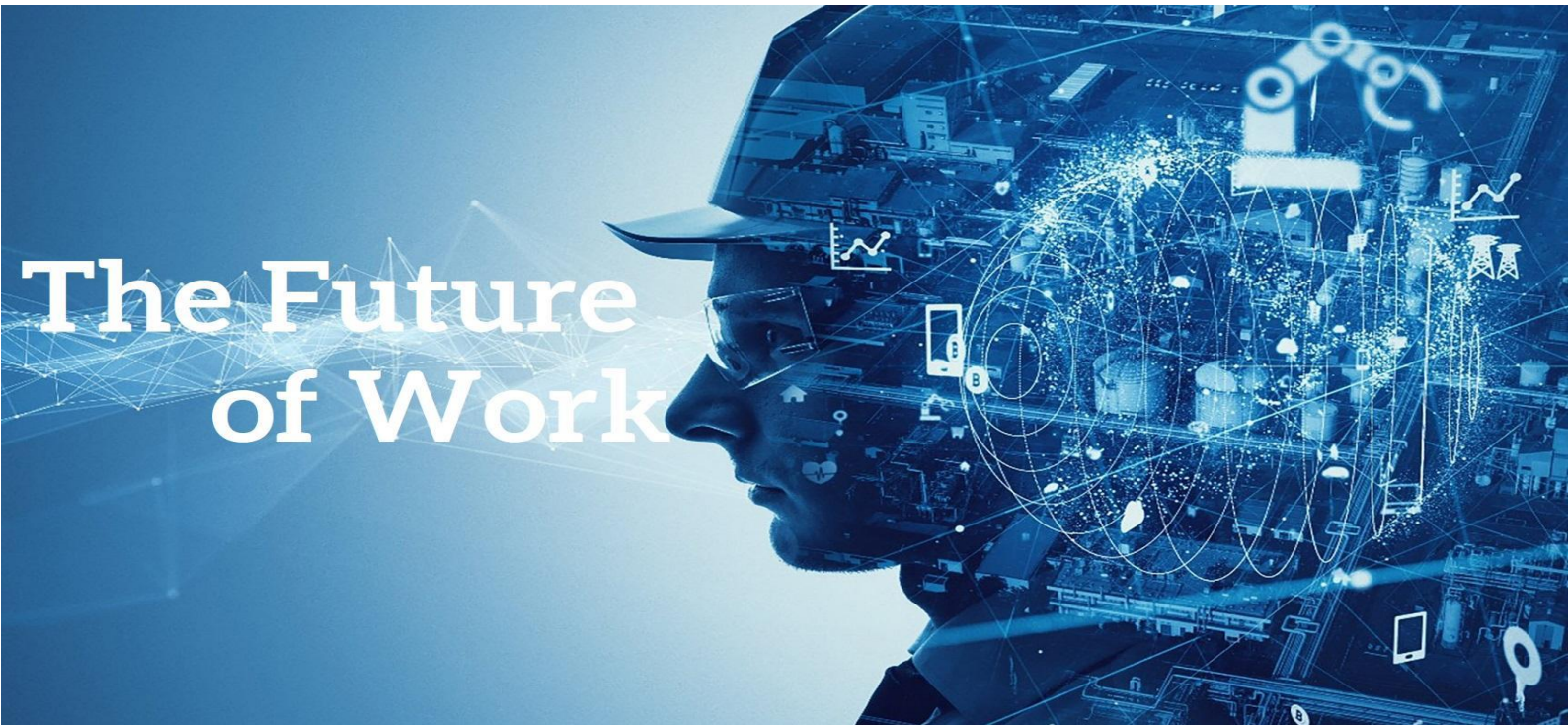
surround= to be all around someone or something on every side
maze= a complicated and confusing arrangement of streets, roads etc
tunnel= a passage that has been dug under the ground for cars, trains etc to go through
robber= someone who steals money or property
mysterious= strange, unexplained, unsolved
discovery= finding, innovation, breakthrough
vessel= a ship or large boat
inscribe= to carefully cut, print, or write words on something, especially on the surface of a stone or coin
archaeologist= someone who studies ancient societies by examining what remains of their buildings, graves, tools etc
represent= to be a symbol of something
excavate= if a scientist or archaeologist excavates an area of land, they dig carefully to find ancient objects, bones etc
honor= respect, pay tribute to
predecessor= someone who had your job before you started doing it
shaft= a passage which goes down through a building or down into the ground, so that someone or something can get in or out
attempt= effort, try, go

precaution= something you do in order to prevent something dangerous or unpleasant from happening
intricate= complicated, complex, sophisticated, tricky
network= system
valuable= things that you own that are worth a lot of money, such as jewellery, cameras etc
overlook= to not notice something, or not see how important it is
thief= someone who steals things from another person or place
astonish= surprise, overwhelm, amaze

exaggeration= a statement or way of saying something that makes something seem better, larger etc than it really is
constitute= to be considered to be something
milestone= a very important event in the development of something
revolutionary= completely new and different, especially in a way that leads to great improvements
archetype= a perfect example of something, because it has all the most important qualities of things that belong to that type

TEST 1

READING PASSAGE 3



According to a **leading** business **consultancy**, 3-14% of

the global **workforce** will need to **switch** to a different **occupation** within the next 10-15 years, and all workers will need to **adapt** as their occupations **evolve** alongside increasingly **capable** machines. **Automation** – or ‘**embodied artificial** intelligence’ (AI) – is one aspect of the **disruptive** effects of technology on the labour market. ‘Disembodied AI’, like the **algorithms** running in our smartphones, is another.

Dr Stella Pachidi from Cambridge Judge Business School believes that some of the most **fundamental** changes are happening as a result of the ‘algorithmic’ of jobs that are dependent on data rather than on production – the so-called knowledge economy. Algorithms are capable of learning from data to **undertake** tasks that previously needed human

leading= best, most important, or most successful
consultancy= a company that gives advice on a particular subject
workforce= all the people who work in a particular industry or company, or are available to work in a particular country or area
switch= to change from doing or using one thing to doing or using another
occupation= job, work, career, profession
adapt= to gradually change your behaviour and attitudes in order to be successful in a new situation
evolve= change, grow, progress, advance
capable= able to do things well
automation= the use of computers and machines instead of people to do a job
embody= represent, exemplify, symbolize
artificial= false, fake, non-natural, man-made
disruptive= causing problems and preventing something from continuing in its usual way
algorithm= a set of instructions that are followed in a fixed order and used for solving a mathematical problem, making a computer program etc

fundamental= important, central, essential, vital
undertake= to accept that you are responsible for a piece of work, and start to do it

judgement, such as reading **legal contracts**, **analysing medical scans** and gathering market intelligence.

‘In many cases, they can **outperform** humans,’ says Pachidi.

‘Organisations are attracted to using algorithms because they want to make choices based on what they consider is “perfect information”, as well as to reduce costs and **enhance productivity**.’

‘But these enhancements are not without **consequences**,’ says Pachidi. ‘If routine **cognitive** tasks are **taken over** by AI, how do professions develop their future **experts**?’ she asks. ‘One way of learning about a job is “legitimate **peripheral** participation” – a **novice** stands next to experts and learns by **observation**. If this isn’t happening, then you need to find new ways to learn.’

Another issue is the extent to which the technology influences or even controls the workforce. For over two years, Pachidi **monitored** a **telecommunications** company. ‘The way telecoms salespeople work is through personal and frequent **contact** with **clients**, using the benefit of experience to **assess** a situation and reach a decision. However, the company had started using a[n] ... algorithm that **defined** when account managers should contact certain customers about which kinds of **campaigns** and what to offer them.’

The algorithm – usually built by **external designers** – often becomes the keeper of knowledge, she explains. In cases like this, Pachidi believes, a **short-sighted** view begins to **creep into** working practices whereby workers learn through the ‘algorithm’s eyes’ and become dependent on its instructions. Alternative **explorations** – where **experimentation** and human **instinct** lead to progress and new ideas – are effectively **discouraged**.

judgement= an opinion that you form, especially after thinking carefully about something
legal= lawful, permissible, legitimate, rightful
contract= an official agreement between two or more people, stating what each will do
analyse= to examine or think about something carefully, in order to understand it
medical= relating to medicine and the treatment of disease or injury
scan= a medical test in which a special machine produces a picture of something inside your body
outperform= to be more successful than someone or something else
enhance= improve, increase, boost
productivity= output, efficiency, production

consequence= result, effect, outcome
cognitive= related to the process of knowing, understanding, and learning something
take over= to take control of something
expert= someone who has a special skill or special knowledge of a subject, gained as a result of training or experience
peripheral= not as important as other things or people in a particular activity, idea, or situation
novice= beginner, learner, trainee, apprentice
observation= the process of watching something or someone carefully for a period of time

monitor= check, watch, supervise, examine
telecommunication= the sending and receiving of messages by telephone, radio, television etc
contact= communication with a person, organization, country etc
client= someone who gets services or advice from a professional person, company, or organization
assess= evaluate, judge, consider
define= to describe something correctly and thoroughly, and to say what standards, limits, qualities etc it has that make it different from other things
campaign= a series of actions intended to achieve a particular result relating to politics or business, or a social improvement

external= outside, exterior, outer
designer= someone whose job is to make plans or patterns for clothes, furniture, equipment etc
short-sighted= not considering the possible effects in the future of something that seems good now – used to show disapproval
creep into= to move in a quiet, careful way, especially to avoid attracting attention
exploration= examination, search, investigation
experimentation= the process of testing various ideas, methods etc to find out how good or effective they are
instinct= a natural tendency to behave in a particular way or a natural ability to know something, which is not learned
discourage= to persuade someone not to do something, especially by making it seem difficult or bad

Pachidi and **colleagues** even observed people developing **strategies** to make the algorithm work to their own advantage. 'We are seeing cases where workers feed the algorithm with false data to reach their **targets**,' she reports.

It's **scenarios** like these that many **researchers** are working to avoid. Their objective is to make AI technologies more **trustworthy** and **transparent**, so that organisations and individuals understand how AI decisions are made. **In the meantime**, says Pachidi, 'We need to make sure we fully understand the **dilemmas** that this new world raises regarding **expertise**, occupational **boundaries** and control.'

Economist Professor Hamish Low believes that the future of work will involve **major transitions** across the whole life course for everyone: 'The traditional **trajectory** of full-time education followed by full-time work followed by a pensioned retirement is a thing of the past,' says Low. Instead, he **envisages** a **multistage** employment life: one where retraining happens across the life course, and where **multiple** jobs and no job happen by choice at different stages.

On the subject of job losses, Low believes the **predictions** are founded on a **fallacy**: 'It assumes that the number of jobs is fixed. If in 30 years, half of 100 jobs are being **carried out** by robots, that doesn't mean we are left with just 50 jobs for humans. The number of jobs will increase: we would **expect** there to be 150 jobs.'

Dr Ewan McGaughey, at Cambridge's Centre for Business Research and King's College London, agrees that 'apocalyptic' views about the future of work are **misguided**. 'It's the laws that **restrict** the **supply** of **capital** to the job market, not the **advent** of new technologies that causes **unemployment**.'

His recently **published** research answers the question of whether automation, AI and robotics will mean a '**jobless** future' by looking at the causes of unemployment. 'History is clear that change can mean **redundancies**. But social policies can **tackle** this through retraining and **redeployment**.'

colleague= coworker, associate, partner, collaborator
strategy= plan, policy, approach, tactic
target= aim, goal, objective

scenario= a situation that could possibly happen
researcher= someone who studies a subject in detail in order to discover new facts or test new ideas
trustworthy= truthful, honest, reliable
transparent= a lie, excuse etc that is transparent does not deceive people
in the meantime= in the period of time between now and a future event, or between two events in the past
dilemma= a situation in which it is very difficult to decide what to do, because all the choices seem equally good or equally bad
expertise= special skills or knowledge in a particular subject, that you learn by experience or training
boundary= the real or imaginary line that marks the edge of a state, country etc, or the edge of an area of land that belongs to someone
major= big, large, considerable, leading
transition= when something changes from one form or state to another
trajectory= the events that happen during a period of time, which often lead to a particular aim or result
envisage= to think that something is likely to happen in the future
multistage= conducted by or occurring in stages
multiple= many, numerous, various

predict= to say that something will happen, before it happens
fallacy= a false idea or belief, especially one that a lot of people believe is true
carry out= to do something that needs to be organized and planned
expect= hope, suppose, think, foresee

apocalyptic= warning people about terrible events that will happen in the future
misguided= intended to be helpful but in fact making a situation worse
restrict= limit, curb, control, constrain
supply= an amount of something that is available to be used
capital= money or property, especially when it is used to start a business or to produce more wealth
advent= arrival, start, beginning
unemployment= when someone does not have a job

publish= to arrange for a book, magazine etc to be written, printed, and sold
jobless= unemployed
redundancy= a situation in which someone has to leave their job, because they are no longer needed
tackle= deal with, work on
redeploy= to move someone or something to a different place or job
(*re*= again i.e rebroadcast)

He adds: 'If there is going to be change to jobs as a result of AI and robotics then I'd like to see governments **seizing the opportunity** to improve policy to **enforce** good job **security**. We can "**reprogramme**" the law to prepare for a fairer future of work and leisure.' McGaughey's findings are **a call to arms** to leaders of organisations, governments and banks to **pre-empt** the coming changes with **bold** new **policies** that **guarantee** full employment, fair incomes and a **thriving** economic **democracy**.

'The promises of these new technologies are **astounding**. They deliver humankind the capacity to live in a way that nobody could have once imagined,' he adds. 'Just as the industrial **revolution** brought people past **subsistence agriculture**, and the **corporate** revolution enabled **mass** production, a third revolution has been **pronounced**. But it will not only be one of technology. The next revolution will be social.'

seize a chance/an opportunity/the initiative= to quickly and eagerly do something when you have the chance to
enforce= to make something happen or force someone to do something
security= things that are done to keep a person, building, or country safe from danger or crime
programme= to arrange for something to happen as part of a series of planned events or activities
a call to arms= something that makes people want to take action and get involved in an attempt to deal with a bad situation
pre-empt= to make what someone has planned to do or say unnecessary or ineffective by saying or doing something first
bold= very strong or bright so that you notice them
policy= a way of doing something that has been officially agreed and chosen by a political party, a business, or another organization
guarantee= ensure, secure, maintain, protect
thriving= a thriving company, business etc is very successful
democracy= a situation or system in which everyone is equal and has the right to vote, make decisions etc

astounding= amazing, surprising, shocking
revolution= a complete change in ways of thinking, methods of working etc
subsistence= the condition of only just having enough money or food to stay alive
agriculture= the practice or science of farming
corporate= shared by or involving all the members of a group
mass= a large amount or quantity of something
pronounced= very great or noticeable

TEST 2

READING PASSAGE 1



The White Horse of Uffington

The cutting of **huge figures** or 'geoglyphs' into the earth of English **hillsides** has **taken place** for more than 3,000 years. There are 56 hill figures **scattered** around England, with the vast **majority** on the chalk **downlands** of the country's southern **counties**. The figures include giants, horses, **crosses** and **regimental badges**. Although the majority of these geoglyphs date within the last 300 years or so, there are one or two that are much older.

huge= giant, enormous, vast, massive
figure= a person in a painting or a model of a person
geoglyph= A large-scale image or design produced in the natural landscape by techniques such as aligning rocks or gravel or removing soil or sod, the complete form of which is visible only aerially or at a distance
hillside= the sloping side of a hill
take place= happen, occur, have effect
scatter= if someone scatters a lot of things, or if they scatter, they are thrown or dropped over a wide area in an irregular way
majority= most of the people or things in a group
downland= gently rolling hill country, especially in southern England
county= an area of a state or country that has its own government to deal with local matters
cross= an object, picture, or mark in the shape of a cross, used as a sign of the Christian faith or for decoration
regimental= connected with a particular regiment (= a large group of soldiers)
badge= a small piece of metal or plastic that you carry to show people that you work for a particular organization

The most famous of these figures is perhaps also the most **mysterious** – the Uffington White Horse in Oxfordshire. The White Horse has recently been **re-dated** and shown to be even older than its **previously assigned ancient** pre-Roman Iron Age date. More **controversial** is the date of the **enigmatic** Long Man of Wilmington in Sussex. While many **historians** are **convinced** the figure is prehistoric, others believe that it was the work of an **artistic monk** from a **nearby priory** and was created between the 11th and 15th centuries.

The **method** of cutting these huge figures was simply to remove the **overlying** grass to **reveal** the **gleaming** white chalk below. However, the grass would soon grow over the geoglyph again unless it was regularly cleaned or **scoured** by a fairly large team of people. One reason that the vast majority of hill figures have disappeared is that when the traditions **associated** with the figures **faded**, people no longer **bothered** or remembered to clear away the grass to expose the chalk outline. Furthermore, over hundreds of years the outlines would sometimes change due to people not always cutting in **exactly** the same place, **thus** creating a different shape to the **original** geoglyph. The fact that any ancient hill figures **survive** at all in England today **is testament to** the strength and **continuity** of local customs and beliefs which, in one case at least, must **stretch** back over **millennia**.

The Uffington White Horse is a **unique, stylised representation** of a horse consisting of a long, **sleek** back, thin **disjointed** legs, a streaming tail, and a bird-like **beaked** head. The **elegant** creature almost **melts into** the **landscape**. The horse is situated 2.5 km from Uffington village on a **steep** slope close to the Late Bronze Age* (c. 7th century BCE)

mysterious= strange, odd, unsolved, inexplicable
re-date= to change the date of
(re-= again i.e rebroadcast)
previously= before, beforehand, formerly, earlier
assign= to give a particular time, value, place etc to something
ancient= antique, old-fashioned, obsolete, outdated, prehistoric
controversial= causing a lot of disagreement, because many people have strong opinions about the subject being discussed
enigmatic= mysterious and difficult to understand
historian= someone who studies history, or the history of a particular thing
convince= to make someone feel certain that something is true
artistic= relating to art or culture
monk= a member of an all-male religious group that lives apart from other people in a monastery
nearby= near, close, in the neighborhood
priory= a building where a group of monks or nuns live, which is smaller and less important than an abbey

method= way, technique, means
overlie= to lie over something
(over-= above; beyond; across i.e overhanging branches, overhead telephone wires)
reveal= tell, disclose, make known, expose
gleaming= bright and shiny from being cleaned
scour= to clean something very thoroughly by rubbing it with a rough material
associated= related, linked, connected
fade= to gradually disappear
bother= to make the effort to do something
exactly= accurately, precisely, correctly
thus= so, therefore, consequently, as a result
original= existing or happening first, before other people or things
survive= to continue to live after an accident, war, or illness
be a testament to something= to prove or show very clearly that something exists or is true
continuity= the state of continuing for a period of time, without problems, interruptions, or changes
stretch= to continue over a period of time or in a series, or to make something do this
millennia= a period of 1,000 years

unique= unusually good and special
stylized= drawn, written, or performed in an artificial style that does not look natural or real, but that is still pleasant to look at
representation= the act of representing someone or something
sleek= sleek hair or fur is straight, shiny, and healthy-looking
disjointed= a disjointed activity or system is one in which the different parts do not work well together
beaked= having or resembling a beak
elegant= beautiful, attractive, or graceful
melt into something= to gradually become hidden by something
landscape= an area of countryside or land of a particular type, used especially when talking about its appearance
steep= a road, hill etc that is steep slopes at a high angle

hillfort of Uffington Castle and below the Ridgeway, a long-distance Neolithic** **track**.

The Uffington Horse is also **surrounded** by Bronze Age burial **mounds**. It is not far from the Bronze Age **cemetery** of Lambourn Seven Barrows, which consists of more than 30 **well-preserved** burial mounds. The **carving** has been placed in such a way as to make it extremely difficult to see from **close quarters**, and like many geoglyphs is best **appreciated** from the air. Nevertheless, there are certain areas of the Vale of the White Horse, the **valley** containing and named after the enigmatic creature, from which an **adequate impression** may be gained. Indeed on a clear day the carving can be seen from up to 30 km away.

The earliest **evidence** of a horse at Uffington is from the 1070s CE when 'White Horse Hill' is mentioned in documents from the nearby Abbey of Abingdon, and the first **reference** to the horse itself is soon after, in 1190 CE. However, the carving is believed to date back much further than that. Due to the similarity of the Uffington White Horse to the stylised **depictions** of horses on 1st century BCE coins, it had been thought that the creature must also date to that period.

However, in 1995 Optically Stimulated Luminescence (OSL) testing was carried out by the Oxford Archaeological Unit on soil from two of the lower layers of the horse's body, and from another cut near the **base**. The result was a date for the horse's **construction** somewhere between 1400 and 600 BCE – in other words, it had a Late Bronze Age or Early Iron Age origin.

The latter end of this date range would tie the carving of the horse in with occupation of the nearby Uffington hillfort, indicating that it may represent a **tribal emblem** marking the land of the **inhabitants** of the hillfort. **Alternatively**, the carving may have been carried out during a Bronze or Iron Age ritual. Some **researchers** see the horse as representing the Celtic*** horse **goddess** Epona, who was **worshipped** as a **protector** of horses, and for her associations with **fertility**. However, the **cult** of Epona was not **imported** from Gaul

track= path, pathway, road, way

be surrounded by something= to be all around someone or something on every side

mound= a pile of earth or stones that looks like a small hill

cemetery= a piece of land, usually not belonging to a church, in which dead people are buried

well-preserved= a well-preserved building or object is old but still in good condition

carving= the activity or skill of carving something

close quarters= if something happens or is done at close quarters, it happens inside a small space or is done from a short distance away

appreciate= to understand how good or useful someone or something is

valley= an area of lower land between two lines of hills or mountains, usually with a river flowing through it

adequate= enough, sufficient

impression= the opinion or feeling you have about someone or something because of the way they seem

evidence= proof, sign, indication

reference= part of something you say or write

in which you mention a person or thing

depiction= description, representation, portrayal

base= the lowest part or surface of something

construction= building, creation

tribal= relating to a tribe or tribes

emblem= symbol, logo, sign, badge

inhabitant= occupant, resident, citizen

alternatively= used for suggesting something different

researcher= someone who studies a subject in detail in order to discover new facts or test new ideas

goddess= a female being who is believed to control the world or part of it, or represents a particular quality

worship= to show respect and love for a god, especially by praying in a religious building

protector= someone or something that protects someone or something else

fertility= the ability of a person, animal, or plant to produce babies, young animals, or seeds

import= to introduce something new or different in a place where it did not previously exist

cult= an extreme religious group that is not part of an established religion

(France) until around the first century CE. This date is at least six centuries after the Uffington Horse was **probably** carved. Nevertheless, the horse had great **ritual** and economic **significance** during the Bronze and Iron Ages, as **attested** by its depictions on **jewellery** and other metal objects. It is possible that the carving represents a goddess in **native mythology**, such as Rhiannon, described in later Welsh mythology as a beautiful woman dressed in gold and riding a white horse.

The fact that geoglyphs can disappear easily, along with their associated rituals and meaning, indicates that they were never intended to be anything more than **temporary gestures**. But this does not lessen their importance. These giant carvings are a fascinating **glimpse** into the minds of their creators and how they viewed the landscape in which they lived.

probably= maybe, possibly, perhaps

ritual= done as part of a rite or ritual

significance= importance, impact

attest= to show or prove that something is true

jewellery= small things that you wear for decoration, such as rings or necklaces

native= your native country, town etc is the place where you were born

mythology= set of ancient myths

temporary= continuing for only a limited period of time

gesture= a movement of part of your body, especially your hands or head, to show what you mean or how you feel

glimpse= a quick look at someone or something that does not allow you to see them clearly

TEST 2

READING PASSAGE 2



Microbes, most of them **bacteria**, have **populated** this planet since long before animal life developed and they will **outlive** us. **Invisible** to **the naked eye**, they are **ubiquitous**. They **inhabit** the soil, air, rocks and water and are present within every form of life, from **seaweed** and **coral** to dogs and humans. And, as Yong explains in his **utterly absorbing** and **hugely** important book, we **mess with** them at our **peril**.

Every **species** has its own **colony** of **microbes**, called a 'microbiome', and these microbes **vary** not only between species but also between individuals and within different parts of each individual. What is amazing is that while the number of

bacteria= very small living things, some of which cause illness or disease
populate= if an area is populated by a particular group of people, they live there
outlive= to remain alive after someone else has died
(out- *being or becoming bigger, further, greater etc than someone or something else i.e outgrow*)
invisible= unseen, unseeable, undetectable
the naked eye= if you can see something with the naked eye, you can see it without using anything to help you, such as a telescope
ubiquitous= seeming to be everywhere – sometimes used humorously
inhabit= live, dwell, occupy, populate
seaweed= a plant that grows in the sea
coral= a hard red, white, or pink substance formed from the bones of very small sea creatures, which is often used to make jewellery
utterly= completely, absolutely, totally, extremely, entirely
absorb= to take in liquid, gas, or another substance from the surface or space around something
hugely= vastly, enormously, immensely, massively
mess with somebody/something= to get involved with someone or something that may cause problems or be dangerous
peril= danger, threat, risk
species= a group of animals or plants whose members are similar and can breed together to produce young animals or plants
colony= a group of animals or plants of the same type that are living or growing together
microbe= an extremely small living thing which you can only see if you use a microscope
vary= differ, diverge, contrast, be different

human **cells** in the average person is about 30 **trillion**, the number of microbial ones is higher – about 39 trillion. At best, Yong **informs** us, we are only 50 per cent human. Indeed, some **scientists** even suggest we should think of each species and its microbes as a **single** unit, **dubbed** a ‘holobiont’.

In each human there are microbes that live only in the stomach, the mouth or the **armpit** and by and large they do so **peacefully**. So ‘bad’ microbes are just microbes out of context. Microbes that sit **contentedly** in the human **gut** (where there are more microbes than there are stars in the galaxy) can become **deadly** if they find their way into the **bloodstream**. These **communities** are **constantly** changing too. The right hand shares just one sixth of its microbes with the left hand. And, of course, we are **surrounded by** microbes. Every time we eat, we **swallow** a million microbes in each gram of food; we are continually swapping microbes with other humans, pets and the world at large.

It’s a fascinating topic and Yong, a young British science journalist, is an **extraordinarily adept** guide. Writing with **lightness** and **panache**, he has a **knack of** explaining **complex** science in **terms** that are both easy to understand and totally **enthralling**. Yong is on a **mission**. Leading us **gently** by the hand, he takes us into the world of microbes – a **bizarre**, **alien** planet – in a **bid to persuade** us to love them as much as he does. By the end, we do.

For most of human history we had no idea that microbes **existed**. The first man to see these extraordinarily **potent** creatures was a Dutch lens-maker called Antony van Leeuwenhoek in the 1670s. Using microscopes of his own design that could **magnify** up to 270 times, he **examined** a drop of water from a **nearby** lake and found it **teeming with tiny** creatures he called ‘animalcules’. It wasn’t until nearly two hundred years later that the research of French biologist Louis Pasteur indicated that some microbes caused disease. It was Pasteur’s ‘germ theory’ that gave bacteria the poor image that **endures** today.

cell= the smallest part of a living thing that can exist independently
trillion= the number 1,000,000,000,000
inform= notify, update, tell
scientist= someone who works or is trained in science
single= only, sole, solo
dub= to give something or someone a name that describes them in some way

armpit= the hollow place under your arm where it joins your body
peacefully= quietly, calmly, tranquilly
contentedly= happy and satisfied because your life is good
gut= all the organs in someone’s body, especially when they have come out of their body
deadly= poisonous, lethal, fatal, toxic
bloodstream= the blood flowing in your body
community= the people who live in the same area, town etc
constantly= continually, continuously, regularly, frequently
be surrounded by something= to be all around someone or something on every side
swallow= to make food or drink go down your throat and towards your stomach

extraordinarily= extremely, very, unusually, amazingly
adept= skillful, skilled, expert, proficient
lightness= the state of being light
panache= a way of doing things that makes them seem easy and exciting, and makes other people admire you
have a knack of doing something= to have a tendency to do something
complex= difficult, complicated
term= a word or expression with a particular meaning, especially one that is used for a specific subject or type of language
enthralling= fascinating, captivating, engrossing
mission= goal, purpose, duty, objective
gently= kindly, smoothly, lightly
bizarre= unusual, odd, strange
alien= unfamiliar, foreign, outlandish
a bid to do something= an attempt to achieve or obtain something
persuade= to make someone decide to do something, especially by giving them reasons why they should do it, or asking them many times to do it

exist= to happen or be present in a particular situation or place
potent= strong, powerful, effective
magnify= to make something seem bigger or louder, especially using special equipment
examine= check, investigate, research, explore
nearby= near, close, close to
teem with somebody/something= to be very full of people or animals, all moving about
tiny= small, little, petite, insignificant
endure= to remain alive or continue to exist for a long time

Yong's book is in many ways a **plea** for microbial **tolerance**, pointing out that while fewer than one hundred species of bacteria bring disease, many thousands more play a vital role in maintaining our health. The book also **acknowledges** that our attitude towards bacteria is not a simple one. We tend to see the dangers posed by bacteria, yet at the same time we are sold yoghurts and drinks that **supposedly nurture** 'friendly' bacteria. In reality, says Yong, bacteria should not be viewed as either friends or **foes**, **villains** or heroes. Instead we should realise we have a **symbiotic** relationship, that can be **mutually** beneficial or mutually **destructive**.

What then do these millions of organisms do? The answer is pretty much everything. New research is now **unravelling** the ways in which bacteria **aid digestion**, **regulate** our immune systems, **eliminate toxins**, produce vitamins, affect our behaviour and even **combat** obesity. 'They actually help us become who we are,' says Yong. But we are facing a growing problem. Our **obsession** with **hygiene**, our overuse of **antibiotics** and our unhealthy, low-fibre diets are **disrupting** the bacterial balance and may be responsible for **soaring** rates of **allergies** and immune problems, such as **inflammatory bowel** disease (IBD).

The most recent research actually turns accepted **norms** upside down. For example, there are studies indicating that the **excessive** use of household **detergents** and antibacterial products actually destroys the microbes that normally keep the more dangerous germs at bay. Other studies show that keeping a dog as a pet gives children early exposure to a diverse range of bacteria, which may help protect them against allergies later.

The readers of Yong's book must be prepared for a decidedly **unglamorous** world. Among the less **appealing** case studies is one about a **fungus** that is **wiping out** entire populations of frogs and that can be **halted** by a rare microbial bacterium. Another is about squid that carry **luminescent** bacteria that

plea= a request that is urgent or full of emotion
tolerance= willingness to allow people to do, say, or believe what they want without criticizing or punishing them
acknowledge= recognize, accept, admit
supposedly= used when saying what many people say or believe is true, especially when you disagree with them
nurture= to feed and take care of a child or a plant while it is growing
foe= an enemy
villain= a bad person or criminal
symbiotic= a symbiotic relationship is one in which the people, organizations, or living things involved depend on each other
mutually= equally, jointly, commonly
destructive= damaging, harmful, detrimental

unravel= solve, find an answer, sort out
aid= help, assist, support
digestion= the process of digesting food
regulate= to make a machine or your body work at a particular speed, temperature etc
immune= someone who is immune to a particular disease cannot catch it
eliminate= remove, eradicate, abolish, exclude, reduce
toxin= a poisonous substance, especially one that is produced by bacteria and causes a particular disease
combat= fight, battle, oppose
obsession= an extreme unhealthy interest in something or worry about something, which stops you from thinking about anything else
hygiene= the practice of keeping yourself and the things around you clean in order to prevent diseases
antibiotic= a drug that is used to kill bacteria and cure infections
disrupt= interrupt, upset, disturb
soar= rise, increase, skyrocket
allergy= a medical condition in which you become ill or in which your skin becomes red and painful because you have eaten or touched a particular substance
inflammatory= an inflammatory disease or medical condition causes inflammation
bowel= one part of this system of tubes

norm= standard, rule, custom
excessive= extreme, too much, unnecessary
detergent= a liquid or powder used for washing clothes, dishes etc

glamorous= attractive, exciting, and related to wealth and success
appealing= interesting, attractive, tempting
fungus= a simple type of plant that has no leaves or flowers and that grows on plants or other surfaces
wipe out= to destroy, remove, or get rid of something completely
halt= stop, pause, finish
luminescence= a soft shining light

protect them against **predators**. However, if you can **overcome** your **distaste** for some of the investigations, the reasons for Yong's **enthusiasm** become clear. The microbial world is a place of wonder. Already, in an attempt to stop mosquitoes **spreading dengue** fever – a disease that **infects** 400 million people a year – mosquitoes are being loaded with a bacterium to block the disease. In the future, our ability to **manipulate** microbes means we could construct buildings with useful microbes built into their walls to fight off infections. Just imagine a **neonatal** hospital ward coated in a specially mixed cocktail of microbes so that babies get the best start in life.

predator= an animal that kills and eats other animals
overcome= to successfully control a feeling or problem that prevents you from achieving something
distaste= dislike, disgust, disfavor
enthusiasm= a strong feeling of interest and enjoyment about something and an eagerness to be involved in it
spread= if something spreads or is spread, it becomes larger or moves so that it affects more people or a larger area
dengue= an illness commonly found in hot countries, caused by the bite of a mosquito which has been infected with a virus
infect= to give someone a disease
manipulate= to make someone think and behave exactly as you want them to, by skillfully deceiving or influencing them
neonatal= relating to babies that have just been born

TEST 2

READING PASSAGE 3

How To Make Wise Decisions

Across cultures, **wisdom** has been considered one of the most **revered** human **qualities**. Although the truly wise may seem **few and far between**, **empirical research** examining wisdom suggests that it isn't an **exceptional** trait **possessed** by a small **handful of bearded philosophers** after all – in fact, the latest studies suggest that most of us have the **ability** to make wise decisions, given the right **context**.

'It appears that **experiential**, situational, and **cultural** factors are even more **powerful** in shaping wisdom than **previously** imagined,' says Associate Professor Igor Grossmann of the University of Waterloo in Ontario, Canada. 'Recent empirical findings from **cognitive**, developmental, social, and personality

wisdom= understanding, knowledge, sense
revered= respected, admired, valued
quality= feature, characteristic, attribute, trait
be few and far between= to be rare
empirical= experiential, experimental, observed
research= study, examination, investigation inquiry
examine= investigate, check, analyze, explore
exceptional= unusually good, outstanding
possess= have, own, hold, keep
handful of= a few, not many, hardly any
beard= facial hair, moustache, mustache
philosopher= theorist, truth-seeker, thinker
ability= aptitude, skill, capability, capacity
context= setting, background, situation, circumstance

experiential= based on experience or related to experience
cultural= belonging or relating to a particular society and its way of life
powerful= influential, controlling, dominant, great
previously= before, beforehand, formerly, earlier
cognitive= reasoning, mental, intellectual

psychology **cumulatively** suggest that people's ability to **reason** wisely **varies dramatically** across experiential and situational contexts. Understanding the role of such **contextual** factors offers **unique insights** into understanding wisdom in daily life, as well as how it can be **enhanced** and taught.'

It seems that it's not so much that some people simply **possess** wisdom and others lack it, but that our ability to reason wisely depends on a variety of **external** factors. 'It is **impossible** to **characterize** thought processes **attributed to** wisdom without considering the **role** of contextual factors,' explains Grossmann. 'In other words, wisdom is not **solely** an "inner quality" but rather **unfolds** as a function of situations people happen to be in. Some situations are more likely to **promote** wisdom than others.'

Coming up with a definition of wisdom is **challenging**, but Grossmann and his **colleagues** have identified four key characteristics as part of a **framework** of wise reasoning. One is **intellectual humility** or **recognition** of the limits of our own knowledge, and another is **appreciation of perspectives** wider than the issue at hand. **Sensitivity** to the possibility of change in social **relations** is also key, along with **compromise** or **integration** of different attitudes and beliefs.

Grossmann and his colleagues have also found that one of the most **reliable** ways to support wisdom in our own day-to-day decisions is to look at **scenarios** from a third-party perspective, as though giving advice to a friend. Research suggests that when adopting a first-person viewpoint we focus on 'the **focal** features of the environment' and when we adopt a third-person, 'observer' viewpoint we reason more broadly and focus more on **interpersonal** and **moral** ideals such as **justice** and **impartiality**. Looking at problems from this more **expansive** viewpoint appears to **foster** cognitive processes related to wise decisions.

What are we to do, then, when **confronted** with situations like a disagreement with a **spouse** or **negotiating a contract** at work, that require us to take a personal **stake**? Grossmann argues that even when we aren't able to change the situation,

psychology= mind, thinking, mindset
cumulatively= in a way that increases by one addition after another
reason= think, rationalize, analyze, solve
vary= differ, diverge, contrast, be different
dramatically= in a great and sudden way
contextual= relating to a particular context
unique= unusually good and special
insight= vision, understanding, awareness
enhance= improve, increase, boost, develop

possess= own, have, hold, enjoy
external= outside, exterior, outward, outer
impossible= not possible, unfeasible, impracticable, unworkable
characterize= describe, portray, illustrate, depict
attribute something to somebody/something= to believe or say that a situation or event is caused by something
role= part, position, responsibility, job
solely= only, merely
inner= internal, innermost, inside, interior
unfold= if a series of events unfolds, they happen
promote= encourage, help, stimulate, support

challenging= demanding, difficult, tough
colleague= coworker, associate, partner, collaborator
framework= structure, frame, scaffold
intellectual= intelligent, knowledgeable, academic
humility= modesty, shyness, self-effacement, unpretentiousness
recognition= identification, detection, distinguishing, differentiation
appreciation= gratitude, gratefulness, obligation, thankfulness
perspective= view, viewpoint, outlook
relation= relative, family member, next of kin
compromise= cooperation, negotiation, concession, conciliation
sensitivity= sympathy, understanding, kindness
integration= addition, mixing, combination, incorporation

reliable= dependable, consistent, unfailing, trustworthy
scenario= a situation that could possibly happen
focal= central, crucial, important, principal
interpersonal= relational, social, personal
moral= good, right, honest, ethical
justice= fairness, impartiality, righteousness, evenhandedness
impartial= neutral, fair, unbiased, objective
expansive= extensive, vast, wide, spread-out
foster= promote, further, advance, cultivate

confront= meet, face, encounter, handle, tackle
spouse= a husband or wife
negotiate= talk, discuss, consult, confer
contract= agreement, bond, indenture
stake= investment, claim, share

we can still **evaluate** these experiences from different perspectives.

For example, in one experiment that took place during the **peak** of a recent economic **recession**, graduating college seniors were asked to **reflect** on their job **prospects**. The students were **instructed** to imagine their career either ‘as if you were a **distant observer**’ or ‘before your own eyes as if you were right there’. Participants in the group **assigned** to the ‘distant observer’ role **displayed** more wisdom-related reasoning (intellectual humility and recognition of change) than did **participants** in the control group.

In another study, couples in long-term romantic relationships were **instructed** to **visualize** an **unresolved** relationship **conflict** either through the eyes of an outsider or from their own perspective. Participants then discussed the **incident** with their partner for 10 minutes, after which they wrote down their thoughts about it. Couples in the ‘other’s eyes’ **condition** were **significantly** more likely to rely on wise reasoning – recognizing others’ perspectives and searching for a **compromise** – compared to the couples in the **egocentric** condition.

‘Ego-decentering promotes greater focus on others and enables a bigger picture, **conceptual** view of the experience, **affording** recognition of intellectual humility and change,’ says Grossmann.

We might **associate** wisdom with intelligence or particular personality traits, but research shows only a small positive relationship between wise thinking and **crystallized** intelligence and the personality **traits** of **openness** and **agreeableness**. ‘It is **remarkable** how much people can vary in their wisdom from one situation to the next, and how much stronger such contextual effects are for understanding the relationship between wise **judgment** and its social and affective **outcomes** as compared to the **generalized** “traits”,’ Grossmann explains. ‘That is, knowing how wisely a person **behaves** in a given situation is more **informative** for understanding their emotions or **likelihood** to **forgive** [or] **retaliate** as compared to knowing whether the person may be wise “in general”.’

evaluate= assess, estimate, calculate, value

peak= the time when something or someone is best, greatest, highest, most successful etc

recession= decline, collapse, downturn, slump

reflect= think, consider, ponder

prospect= possibility, likelihood, probability, potential

distant= far, remote, faraway

instruct= teach, train, coach, tutor, educate

observer= spectator, witness, viewer, onlooker

assign= attribute, ascribe, impute

display= show, exhibition, presentation, demonstration

participant= member, contributor, partaker

instruct= to officially tell someone what to do

visualize= to form a picture of someone or something in your mind

unresolved= an unresolved problem or question has not been answered or solved

conflict= a state of disagreement or argument between people, groups, countries etc

incident= an event, especially one that is unusual, important, or violent

condition= state, form, situation, circumstance

significantly= considerably, notably, substantially

compromise= cooperation, negotiation, concession

egocentric= thinking only about yourself and not about what other people might need or want

conceptual= dealing with ideas, or based on them

afford= to provide something or allow something to happen

associate= connect, relate, link, correlate

crystallized=

trait= a particular quality in someone’s character

openness= honesty, directness, frankness, sincerity

agreeableness= friendliness, kindness, sociability, pleasantness

remarkable= notable, amazing, outstanding, extraordinary

judgment= an opinion that you form, especially after thinking carefully about something

generalized= global, universal, widespread, sweeping, comprehensive

outcome= result, consequence, effect, conclusion

behave= act, perform, work, deport yourself

informative= educational, revealing, enlightening, useful, instructive

likelihood= possibility, probability, prospect, chance

forgive= pardon, excuse, absolve, exonerate, let off

retaliate= react, hit back, strike back, get even, get revenge

TEST 3

READING PASSAGE 1



Roman Shipbuilding And Navigation

Shipbuilding today is **based on** science and ships are

built using computers and **sophisticated tools**. Shipbuilding in **ancient Rome**, however, was more of an art **relying on estimation, inherited techniques** and personal experience. The Romans were not **traditionally sailors** but mostly land-based people, who learned to build ships from the people that they **conquered**, namely the Greeks and the Egyptians.

There are a few **surviving written documents** that give **descriptions** and **representations** of ancient Roman ships, including the sails and **rigging**. **Excavated vessels** also provide some **clues** about ancient shipbuilding techniques. Studies of these have taught us that ancient Roman shipbuilders built the **outer hull** first, then **proceeded** with the **frame** and the rest of the ship. **Planks** used to build the outer

base on= to use something as the thing from which something else is developed
sophisticated= complex, complicated, difficult
tool= instrument, implement, device, means
ancient= antique, old-fashioned, obsolete, outdated, prehistoric
rely on= depend on, count on, trust, be sure of
estimation= approximation, estimate, assessment, valuation
inherit= receive, get, come into, accede to
technique= method, way, means
traditionally= according to tradition
sailor= someone who works on a ship
conquer= defeat, beat, overpower

survive= live, endure, continue, last, stay alive
document= text, file, paper, record
description= account, report, explanation, portrayal
representation= symbol, image, depiction, demonstration
rigging= ropes, chains, wires
excavate= dig, mine, quarry, exhume
vessel= a ship or large boat
clue= sign, hint, evidence
outer= outside, exterior, outdoor, outward
hull= the main part of a ship that goes in the water
proceed= continue, keep, go on
frame= structure, framework, scaffold, support
plank= a long narrow piece of wooden board, used especially for making structures to walk on

hull were **initially sewn** together. Starting from the 6th century BCE, they were **fixed** using a method called **mortise and tenon**, whereby one plank locked into another without the need for **stitching**. Then in the first centuries of the current **era**, Mediterranean shipbuilders **shifted** to another shipbuilding method, still in use today, which **consisted of** building the frame first and then proceeding with the hull and the other **components** of the ship. This method was more **systematic** and **dramatically** shortened ship **construction** times. The ancient Romans built large **merchant** ships and warships whose size and technology were unequalled until the 16th century CE.

Warships were built to be **lightweight** and very **speedy**. They had to be able to sail near the **coast**, which is why they had no **ballast** or **excess load** and were built with a long, **narrow** hull. They did not **sink** when damaged and often would lie **crippled** on the sea's **surface** following **naval battles**. They had a **bronze battering ram**, which was used to **pierce** the **timber** hulls or break the **oars** of **enemy** vessels. Warships used both wind (sails) and human power (oarsmen) and were therefore very fast. Eventually, Rome's navy became the largest and most **powerful** in the Mediterranean, and the Romans had control over what they therefore called Mare Nostrum meaning 'our sea'.

There were many kinds of warship. The 'trireme' was the **dominant** warship from the 7th to 4th century BCE. It had **rowers** in the top, middle and lower levels, and **approximately** 50 rowers in each bank. The rowers at the bottom had the most uncomfortable position as they were under the other rowers and were **exposed** to the water entering through the oar-holes. It is worth noting that **contrary** to popular **perception**, rowers were not slaves but mostly Roman citizens **enrolled** in the military. The trireme was **superseded** by larger ships with even more rowers.

initially= firstly, at first, primarily
sew= stitch, seam, baste, hem
fix= repair, mend, correct
mortise= a hole cut in a piece of wood or stone so that the shaped end of another piece will fit there firmly
tenon= an end of a piece of wood, that has been cut to fit exactly into a mortise in order to form a strong joint
stitching= sewing, seam, needlework, embroidery
era= age, epoch, eon, period
shift= change, alter, transfer
consist of something= be made of, be made up of, contain, be composed of
component= part, piece, element
systematic= organized carefully and done thoroughly
dramatically= radically, noticeably, severely, considerably, spectacularly, vividly
construction= building, creation, erection
merchant= seller, trader, tradesperson

lightweight= trivial, insubstantial, inconsequential, unimportant
speedy= quick, immediate, fast
coast= shore, shoreline, coastline, seashore
ballast= heavy material that is carried by a ship to make it more steady in the water
excess= extra, spare, surplus
load= weight, cargo, shipment, capacity
narrow= thin, fine, slim, slender, slight
sink= descend, drop, go under, go down, go under the surface
cripple= to damage something badly so that it no longer works or is no longer effective
surface= the top layer of an area of water or land
naval= marine, nautical, maritime, seafaring
battle= fight, clash, combat, encounter
bronze= a hard metal that is a mixture of copper and tin
battering= when someone or something is severely damaged, defeated, criticized etc
ram= a machine that hits something again and again to force it into a position
pierce= stab, impale, cut, slice
timber= wood used for building or making things
oar= a long pole with a wide flat blade at one end, used for rowing a boat
enemy= opponent, adversary, foe, rival
powerful= influential, controlling, dominant, great

dominant= more powerful, important, or noticeable than other people or things
rower= oarsperson, sculler, cockswain
approximately= about, around, roughly, almost
expose= subject, endanger, imperil, put in danger
contrary= conflicting, opposing, different, disagreeing
perception= view, opinion
supersede= succeed, supplant, replace, surpass
enroll= register, join, sign up

Merchant ships were built to **transport** lots of **cargo** over long distances and at a **reasonable** cost. They had a wider hull, double planking and a solid interior for added **stability**. Unlike warships, their V-shaped hull was deep underwater, meaning that they could not sail too close to the coast. They usually had two huge side **rudders** located off the **stern** and controlled by a small **tiller** bar connected to a system of cables. They had from one to three **masts** with large square sails and a small triangular sail at the bow. Just like warships, merchant ships used oarsmen, but coordinating the hundreds of rowers in both types of ship was not an easy task. In order to assist them, music would be played on an instrument, and oars would then keep time with this.

The cargo on merchant ships included raw materials (e.g. iron bars, copper, marble and **granite**), and agricultural products (e.g. grain from Egypt's Nile valley). During the Empire, Rome was a huge city by ancient **standards** of about one million **inhabitants**. Goods from all over the world would come to the city through the port of Pozzuoli situated west of the bay of Naples in Italy and through the **gigantic** port of Ostia situated at the mouth of the Tiber River. Large merchant ships would **approach** the destination port and, just like today, be **intercepted** by a number of towboats that would **drag** them to the **quay**.

The time of travel along the many sailing routes could vary widely. **Navigation** in ancient Rome did not rely on sophisticated instruments such as **compasses** but on experience, local knowledge and **observation** of natural **phenomena**. In conditions of good **visibility**, seamen in the Mediterranean often had the mainland or islands in sight, which greatly **facilitated** navigation. They sailed by noting their **position** relative to a **succession** of **recognisable** landmarks. When weather conditions were not good or where land was no longer visible, Roman **mariners** estimated directions from the pole star or, with less **accuracy**, from the Sun at noon. They also estimated directions relative to the wind and **swell**. Overall, shipping in ancient Roman times **resembled** shipping today with large vessels **regularly** crossing the seas and bringing **supplies** from their Empire.

transport= convey, move, bring, carry, ship
cargo= load, freight, consignment, shipment
reasonable= inexpensive, affordable, cheap, moderate, economical
stability= constancy, steadiness, firmness, solidity
rudder= a flat part at the back of a ship or aircraft that can be turned in order to control the direction in which it moves
stern= the back of a ship
tiller= wheel, rudder, controls
mast= a tall pole on which the sails or flags on a ship are hung

standard= norm, average, benchmark
granite= a very hard grey rock, often used in building
inhabitant= occupant, resident, citizen
gigantic= huge, enormous, vast
approach= come near, move toward
intercept= interrupt, stop, seize, capture
drag= pull, haul, draw, heave
quay= dock, dockside, wharf, pier, harbor

navigation= steering, direction finding, routing
compass= an instrument that shows directions and has a needle that always points north
observation= watching, scrutiny, inspection
phenomenon= occurrence, fact, experience, happening
visibility= distance, range, horizon
facilitate= help, aid, assist, make easy
position= location, place, site, spot
succession= series, sequence, chain, run
recognizable= familiar, identifiable, detectable, detectable, distinguishable, noticeable
mariner= a sailor
accuracy= correctness, accurateness, exactness, precision
swell= the way the sea moves up and down
resemble= look like, bear a resemblance to, be similar to
regularly= frequently, often, repeatedly, recurrently
supply= source, stock, amount, quantity, resource

TEST 3

READING PASSAGE 2

Climate Change Reveals Ancient Artefacts In Norway's Glaciers

Well above the treeline in Norway's highest mountains,

ancient fields of ice are **shrinking** as Earth's climate warms. As the ice has **vanished**, it has been giving up the **treasures** it has **preserved** in cold storage for the last 6,000 years – items such as ancient **arrows** and skis from Viking Age traders. And those **artefacts** have provided **archaeologists** with some **surprising insights** into how ancient Norwegians made their livings.

B Organic materials like **textiles** and **hides** are relatively **rare** finds at archaeological sites. This is because unless they're protected from the **microorganisms** that cause decay, they tend not to last long. **Extreme** cold is one **reliable** way to keep artefacts relatively fresh for a few thousand years, but once **thawed** out, these materials experience **degradation** relatively **swiftly**. With climate change shrinking ice cover around the

ancient= antique, old-fashioned, obsolete, outdated, prehistoric

shrink= to become smaller, or to make something smaller, through the effects of heat or water

vanish= disappear, go, evaporate

treasure= a group of valuable things such as gold, silver, jewels etc

preserve= protect, conserve, safeguard, save

arrow= a weapon usually made from a thin straight piece of wood with a sharp point at one end, that you shoot with a bow

artefact= object, article, item, piece

archaeologist= someone who studies ancient societies by examining what remains of their buildings, graves, tools etc

surprising= astonishing, astounding, amazing, shocking, startling

insight= vision, understanding, awareness, perception

textile= fabric, cloth, material, knit

hide= skin, pelt, fleece, fur

rare= uncommon, unusual, odd

microorganism= bug, germ, virus, microbe, bacteria

extreme= great, tremendous, severe, acute, intense

reliable= dependable, unswerving, unfailing, trustworthy

thaw= melt, defrost, soften, liquify

degradation= an experience or situation that makes you feel ashamed and angry

swift= speedy, fast, quick, rapid

world, **glacial** archaeologists need to **race** the clock to find newly **revealed** artefacts, **preserve** them, and study them. If something **fragile** dries and is windblown it might very soon be lost to science, or an **arrow** might be **exposed** and then covered again by the next snow and remain **well-preserved**. The **unpredictability** means that glacial archaeologists have to be **systematic** in their **approach** to fieldwork.

C Over a nine-year period, a team of archaeologists, which included Lars Pilo of Oppland County Council, Norway, and James Barrett of the McDonald Institute for Archaeological Research, **surveyed patches** of ice in Oppland, an area of south-central Norway that is home to some of the country's highest mountains. **Reindeer** once **congregated** on these icy patches in the later summer months to **escape** biting insects, and from the late Stone Age**, hunters followed. In addition, trade **routes threaded** through the mountain passes of Oppland, linking **settlements** in Norway to the rest of Europe. The slow but **steady** movement of glaciers tends to **destroy** anything at their **bases**, so the team focused on stationary patches of ice, mostly above 1,400 metres. That ice is found **amid** fields of frost-weathered **boulders**, fallen rocks, and exposed **bedrock** that for nine months of the year is **buried beneath** snow. 'Fieldwork is hard work – hiking with all our equipment, often camping on **permafrost** – but very **rewarding**. You're rescuing the archaeology, bringing the melting ice to wider attention, discovering a **unique** environmental history and really connecting with the natural environment,' says Barrett.

D At the **edges** of the **contracting** ice patches, archaeologists found more than 2,000 artefacts, which formed a material **record** that ran from 4,000 BCE to the beginnings of the Renaissance in the 14th century. Many of the artefacts are **associated** with hunting. Hunters would have easily **misplaced** arrows and they often **discarded** broken **bows** rather than take them all the way home. Other items could have been used by hunters **traversing** the high mountain passes of Oppland: all-purpose items like tools, skis, and horse **tack**.

E Barrett's team radiocarbon-dated 153 of the artefacts and compared those dates to the **timing** of major environmental changes in the **region** – such as periods of cooling or warming – and major social and economic **shifts** – such as the growth

glacial= relating to ice and glaciers, or formed by glaciers
race= run, sprint, hurry, speed, dash, rush
reveal= expose, uncover, show, bare
preserve= protect, conserve, safeguard, save
fragile= easily broken or damaged
arrow= a weapon usually made from a thin straight piece of wood with a sharp point at one end, that you shoot with a bow
exposed= not covered
well-preserved= a well-preserved building or object is old but still in good condition
unpredictable= random, erratic, changeable, impulsive, volatile, irregular, variable
systematic= orderly, methodical, regular, organized
approach= method, tactic, line, slant, style

survey= examine, review, study, inspect, investigate
patch= area, spot, blotch, bit, smear
reindeer= a large deer with long wide antlers (=horns), that lives in cold northern areas
congregate= to come together in a group
escape= flee, bolt, abscond, run away, get away
route= way, road, course, path, direction
thread= to put a thread, string, rope etc through a hole
settlement= an official agreement or decision that ends an argument, a court case, or a fight, or the action of making an agreement
steady= stable, firm, fixed, solid
destroy= damage, break, spoil, wreck, ruin
base= the lowest part or surface of something
amid= among, amongst, within, in
boulder= a large round piece of rock
bedrock= base, basis, core, heart, root
bury= to put someone who has died in a grave
beneath= under, underneath, below
permafrost= a layer of soil that is always frozen in countries where it is very cold
rewarding= satisfying, worthwhile, gratifying, pleasing, fulfilling
unique= unusually good and special

edge= brink, verge, threshold, point
contract= to become smaller or narrower
record= note, memo, document, information
be associated with somebody or something= to be related to a particular subject, activity etc
misplace= to lose something for a short time by putting it in the wrong place
discard= to get rid of something
bow= a weapon used for shooting arrows, made of a long thin piece of wood held in a curve by a tight string
traverse= cross, pass over, get over
tack= a small nail with a sharp point and a flat top

timing= the skill of doing something at exactly the right time
region= area, district, county, section
shift= change, alter, transfer

of farming **settlements** and the **spread** of international trade **networks** leading up to the Viking Age. They found that some periods had produced lots of artefacts, which **indicates** that people had been pretty active in the mountains during those times. But there were few or no signs of activity during other periods.

F What was **surprising**, according to Barrett, was the timing of these periods. Oppland's mountains present **daunting terrain** and in periods of extreme cold, glaciers could **block** the higher mountain **passes** and make travel in the upper reaches of the mountains extremely difficult. Archaeologists **assumed** people would **stick** to lower **elevations** during a time like the Late Antique Little Ice Age, a short period of deeper-than-usual cold from about 536-600 CE. But it turned out that hunters kept regularly **venturing** into the mountains even when the climate turned cold, based on the amount of **stuff** they had **apparently** dropped there. '**Remarkably**, though, the finds from the ice may have continued through this period, perhaps suggesting that the importance of mountain hunting increased to **supplement** failing agricultural **harvests** in times of low **temperatures**,' says Barrett. A colder turn in the Scandinavian climate would likely have meant **widespread** crop **failures**, so more people would have **depended on** hunting to **make up for** those losses.

G Many of the artefacts Barrett's team **recovered** date from the beginning of the Viking Age, the 700s through to the 900s CE. **Trade** networks connecting Scandinavia with Europe and the Middle East were **expanding** around this time. Although we usually think of ships when we think of Scandinavian expansion, these recent **discoveries** show that **plenty of** goods travelled on **overland** routes, like the mountain passes of Oppland. And growing Norwegian towns, along with **export** markets, would have created a **booming demand** for hides to fight off the cold, as well as **antlers** to make useful things like combs. Business must have been good for hunters.

H Norway's mountains are probably still hiding a lot of history – and **prehistory** – in **remote** ice patches. When Barrett's team looked at the dates for their **sample** of 153 artefacts, they noticed a gap with almost no artefacts from about 3,800 to 2,200 BCE. In fact, archaeological finds from that period are **rare** all over Norway. The researchers say that could be because many of those artefacts have already **disintegrated** or are still frozen in the ice. That means archaeologists could be **extracting** some of those artefacts from **retreating** ice in years to come.

settlement= community, village, town, neighborhood
spread= expanse, distribution, range, extent, increase
network= system
indicate= specify, show, signpost, direct, point to

surprising= astonishing, astounding, amazing, shocking, startling
daunting= deterring, discouraging, scaring, frightening
terrain= a particular type of land
block= stop, obstruct, impede, hinder, jam, prevent
pass= passage, route, road, way
assume= guess, think, suppose, presume
stuff= gear, equipment, property, kit
stick= attach, glue, fix, join
elevation= a height above the level of the sea
venture= a new business activity that involves taking risks
apparently= actually, evidently, obviously
remarkably= extraordinarily, outstandingly, extremely
supplement= addition, extra, complement, enhancement
harvest= the time when crops are gathered from the fields, or the act of gathering them
temperature= a measure of how hot or cold a place or thing is
widespread= extensive, prevalent, general, common, rife
failure= an occasion when crops do not grow or produce food, for example because of bad weather
depend on= rely on, count on, bank on, trust
make up for= compensate

recover= replace something that has been lost or to get better after an illness, accident, shock etc
trade= the activity of buying, selling, or exchanging goods within a country or between countries
expand= enlarge, get bigger, develop
discovery= finding, innovation, breakthrough
plenty of= a lot of, lots of
overland= across land, not by sea or air
export= the business of selling and sending goods to other countries
booming= having a period of great prosperity or rapid economic growth
demand= request, plea, call
antler= one of the two horns of a male deer

prehistory= early history, dawn of time, ancient history
remote= distant, isolated, far-flung, far-off
sample= example, model, illustration
disintegrate= to break up, or make something break up, into very small pieces
extract= to remove an object from somewhere, especially with difficulty
retreat= if an area of water, snow, or land retreats, it gradually gets smaller

TEST 3

READING PASSAGE 3

Plant 'Thermometer' Triggers Springtime Growth By Measuring Night-time Heat

A An **international** team of scientists led by the University of Cambridge has **discovered** that the 'thermometer' **molecule** in plants enables them to develop according to **seasonal temperature** changes. **Researchers** have **revealed** that molecules called **phytochromes** – used by plants to **detect** light during the day – actually change their **function** in darkness to become **cellular** temperature **gauges** that measure the heat of the night. The new findings, published in the journal *Science*, show that **phytochromes** control **genetic switches** in response to temperature as well as light to **dictate** plant development.

B At night, these molecules change states, and the **pace** at which they change is 'directly **proportional** to temperature', say scientists, who compare phytochromes to **mercury** in a

international= global, worldwide, universal, transnational
discover= find, uncover, realize
thermometer= a piece of equipment that measures the temperature of the air, of your body etc
molecule= particle, bit, iota, jot
seasonal= periodic, cyclic, regular, recurrent, cyclical
temperature= a measure of how hot or cold a place or thing is
researcher= someone who studies a subject in detail in order to discover new facts or test new ideas
reveal= tell, disclose, make known, expose
detect= see, witness, spot
function= purpose, meaning, role, job
cellular= consisting of or relating to the cells of plants or animals
gauge= an instrument for measuring the size or amount of something
phytochrome= any of a group of proteins bound to light-absorbing pigments in many plants that play a role in initiating floral and developmental processes when activated by red or near-infrared radiation
genetic= relating to genes or genetics
switch= change, shift, adjustment, difference, modification
dictate= determine, influence, shape, control
pace= speed, rapidity, rate
proportional= relative, relational, comparative
mercury= a heavy silver-white poisonous metal that is liquid at ordinary temperatures, and is used in thermometers

thermometer. The warmer it is, the faster the molecular change – **stimulating** plant **growth**.

C Farmers and gardeners have known for hundreds of years how **responsive** plants are to temperature: warm winters cause many trees and flowers to **bud** early, something humans have long used to **predict** weather and harvest times for the coming year. The latest research **pinpoints** for the first time a molecular **mechanism** in plants that **reacts** to temperature – often **triggering** the buds of spring we long to see at the end of winter.

D With weather and temperatures set to become ever more **unpredictable** due to climate change, researchers say the discovery that this light-sensing molecule also functions as the **internal** thermometer in plant cells could help us **breed** tougher crops. 'It is **estimated** that agricultural **yields** will need to double by 2050, but climate change is a major threat to **achieving** this. Key crops such as wheat and rice are **sensitive** to high temperatures. **Thermal** stress reduces crop yields by around 10% for every one degree increase in temperature,' says lead researcher Dr Philip Wigge from Cambridge's Sainsbury Laboratory. 'Discovering the molecules that allow plants to **sense** temperature has the **potential** to **accelerate** the breeding of crops **resilient** to thermal stress and climate change.'

E In their active state, phytochrome molecules **bind** themselves to DNA to **restrict** plant growth. During the day, sunlight activates the molecules, slowing down growth. If a plant finds itself in **shade**, phytochromes are quickly inactivated – enabling it to grow faster to find sunlight again. This is how plants **compete** to **escape** each other's shade. 'Light-driven changes to phytochrome activity **occur** very fast, in less than a second,' says Wigge. At night, however, it's a different story. Instead of a rapid **deactivation** following **sundown**, the molecules **gradually** change from their active to inactive state. This is called 'dark reversion'. 'Just as mercury rises in a thermometer, the rate at which phytochromes **revert** to their inactive state during the night is a direct **measure** of temperature,' says Wigge.

F 'The lower the temperature, the slower the rate at which phytochromes revert to **inactivity**, so the molecules spend more time in their active, **growth-suppressing** state. This is why plants are slower to grow in winter. Warm temperatures **accelerate** dark reversion, so that phytochromes rapidly reach an inactive state and **detach** themselves from the plant's DNA – allowing genes to be expressed and plant growth to **resume**.' Wigge believes phytochrome thermo-sensing **evolved** at a later stage, and **co-opted** the biological network already used for light-based growth during the downtime of night.

stimulate= excite, inspire, motivate, encourage
arouse
growth= development, evolution, progress

responsive= reacting quickly, in a positive way
bud= to produce buds
predict= to say that something will happen, before it happens
pinpoint= identify, locate, find
mechanism= means, method, system, procedure
react= respond, counter, answer, reply
trigger= activate, cause, start, initiate

unpredictable= random, erratic, changeable, impulsive, volatile, irregular, variable
internal= interior, inner, inside, core
breed= have babies, reproduce, procreate
estimate= guess, reckon, value, appraise, guesstimate
yield= harvest, crop, produce
achieve= attain, realize, reach, complete, do
sensitive= delicate, irritable, susceptible, allergic
thermal= relating to or caused by heat
sense= detect, identify, recognize, feel
potential= the possibility that something will develop in a particular way, or have a particular effect
accelerate= speed up, hurry, quicken
resilient= hardy, strong, tough, robust, resistant

bind= attach, connect, unite, tie
restrict= limit, curb, control, constrain
occur= take place, happen, have effect
shade= shadow, dark, darkness
compete= try to win, contend, fight
sundown= sunset
escape= flee, bolt, abscond, run away, get away
gradually= slowly, steadily, in stages
revert= to change back to a situation that existed in the past
measure= amount, degree, quantity, portion
deactivate= neutralize, disable, disengage, switch off

inactivity= the state of not doing anything, not moving, or not working
suppress= overpower, overwhelm, overturn, conquer, defeat
detach= separate, remove, disengage, disconnect, isolate
evolve= change, grow, progress, advance
co-opt= to persuade someone to help or support you
accelerate= hurry, hasten, quicken, rush
resume= restart, continue, start again

G Some plants mainly use day length as an **indicator** of the season. Other **species**, such as **daffodils**, have **considerable** temperature sensitivity, and can flower months **in advance** during a warm winter. In fact, the discovery of the **dual** role of phytochromes provides the science behind a **well-known rhyme** long used to **predict** the coming season: **oak** before **ash** we'll have a **splash**, ash before oak we're in for a **soak**. Wigge explains: 'Oak trees rely much more on temperature, likely using phytochromes as thermometers to dictate development, whereas ash trees **rely on** measuring day length to **determine** their seasonal timing. A warmer spring, and **consequently** a higher **likeliness** of a hot summer, will result in oak leafing before ash. A cold spring will see the opposite. As the British know only too well, a colder summer is likely to be a rain-soaked one.'

H The new **findings** are the **culmination** of twelve years of research involving scientists from Germany, Argentina and the US, as well as the Cambridge team. The work was done in a model system, using a **mustard** plant called Arabidopsis, but Wigge says the phytochrome genes necessary for temperature sensing are found in crop plants as well. 'Recent **advances** in plant genetics now mean that scientists are able to **rapidly identify** the genes controlling these **processes** in crop plants, and even **alter** their activity using **precise** molecular "scalpels",' adds Wigge. 'Cambridge is **uniquely well-positioned** to do this kind of research as we have **outstanding collaborators nearby** who work on more applied aspects of plant **biology**, and can help us **transfer** this new knowledge into the field.'

indicator= sign, marker, guide, statistic
species= a group of animals or plants whose members are similar and can breed together to produce young animals or plants
dual= having two of something or two parts
well-known= known by a lot of people
rhyme= a short poem or song, especially for children, using words that rhyme
daffodil= a tall yellow spring flower with a tube-shaped part in the middle
considerable= large, major, big, significant
in advance= before something happens or is expected to happen
predict= forecast, foresee, envisage, expect
oak= a large tree that is common in northern countries, or the hard wood of this tree
ash= the soft grey powder that remains after something has been burned
splash= the sound of a liquid hitting something or being moved around quickly
soak= wet, sodden, drench, douse, douse
rely on/upon somebody/something= depend on, count on, trust
determine= decide, conclude, establish, finalize
consequently= so, thus, therefore, accordingly, as a result
likeliness= likelihood

finding= discovery, conclusion, result, outcome
the culmination of something= something, especially something important, that happens at the end of a long period of effort or development
mustard= a plant with yellow flowers and seeds that are used to make mustard sauce
advance= development, improvement, spread progress
rapidly= fast, quickly, speedily, swiftly
identify= classify, recognize, find, detect
alter= change, modify, adjust, vary
precise= exact, accurate, correct
uniquely= exclusively, exceptionally, distinctively
outstanding= unusually good, exceptional
transfer= move, transport, relocate, remove, shift
nearby= near, close, in the neighborhood
well-positioned= to be in a situation in which you will be able to do something successfully
collaborator= coworker, colleague, partner teammate
biology= the scientific study of living things
process= procedure, development, course, progression

TEST 4

READING PASSAGE 1

Roman Tunnels

The **Persians**, who lived in present-day Iran, were one of the first **civilizations** to build **tunnels** that provided a **reliable** supply of water to human **settlements** in dry areas. In the early first **millennium** BCE, they introduced the **qanat method** of tunnel **construction**, which **consisted of** placing posts over a hill in a straight line, to ensure that the tunnel kept to its **route**, and then digging **vertical shafts** down into the ground **at regular intervals**. **Underground**, workers removed the earth from between the ends of the shafts, creating a tunnel. The **excavated** soil was taken up to the **surface** using the shafts, which also provided **ventilation** during the work. Once the tunnel was completed, it allowed water to **flow** from the top of a hillside down towards a **canal**, which **supplied** water for human use. **Remarkably**, some qanats built by the Persians 2,700 years ago are still in use today.

persian= someone from Iran, especially in the time when it was called Persia
civilization= people society, nation, culture
tunnel= channel, passageway, subway, shaft
reliable= dependable, consistent, steadfast, unflinching
settlement= community, village, town, neighborhood
millennium= a period of 1,000 years
qanat= an underground aqueduct used (as in the Middle East) to convey water from a source (such as an aquifer) through the force of gravity especially for the purpose of irrigation
method= way, technique, means
construction= building, creation
consist of something= be made of, be made up of, contain, be composed of
route= pass, passage, road, way
vertical= perpendicular, upright, erect, straight up ≠ horizontal
shaft= a passage which goes down through a building or down into the ground, so that someone or something can get in or out
at regular intervals= something that happens at regular intervals happens often
underground= below the surface of the earth
excavate= dig, mine, quarry, exhume
surface= outside, exterior, top
ventilate= air, air out, freshen, circulate air
flow= flood, stream, gush, run
canal= a long passage dug into the ground and filled with water, either for boats to travel along, or to take water to a place
supply= provide, bring, give

They later **passed on** their knowledge to the Romans, who also used the qanat method to construct water-supply tunnels for **agriculture**. Roman qanat tunnels were constructed with vertical shafts dug at intervals of between 30 and 60 meters. The shafts were **equipped** with **handholds** and **footholds** to help those climbing in and out of them and were covered with a wooden or stone **lid**. To **ensure** that the shafts were vertical, Romans hung a **plumb** line from a **rod** placed across the top of each shaft and made sure that the weight at the end of it hung in the center of the shaft. Plumb lines were also used to **measure** the depth of the shaft and to **determine** the **slope** of the tunnel. The 5.6-kilometer-long Claudius tunnel, built in 41 CE to **drain** the Fucine Lake in central Italy, had shafts that were up to 122 meters deep, took 11 years to build and involved approximately 30,000 workers.

By the 6th century BCE, a second method of tunnel construction **appeared** called the counter-excavation method, in which the tunnel was constructed from both **ends**. It was used to cut through high mountains when the qanat method was not a practical **alternative**. This method required greater planning and **advanced** knowledge of surveying, mathematics and **geometry** as both ends of a tunnel had to meet correctly at the center of the mountain. **Adjustments** to the **direction** of the tunnel also had to be made whenever builders **encountered** geological problems or when it **deviated** from its set path. They **constantly** checked the tunnel's advancing direction, for example, by looking back at the light that **penetrated** through the tunnel mouth, and made corrections whenever necessary. Large deviations could happen, and they could result in one end of the tunnel not being **usable**. An **inscription** written on the side of a 428-meter tunnel, built by the Romans as part of the Saldae **aqueduct** system in modern-day Algeria, describes how the two teams of builders missed each other in the mountain and how the later construction of a **lateral** link between both **corridors** corrected the **initial error**.

The Romans dug tunnels for their roads using the counter-excavation method, whenever they encountered **obstacles** such as hills or mountains that were too high for roads to pass over. An example is the 37-meter-long, 6-meter-high, Furlo Pass Tunnel built in Italy in 69-79 CE. Remarkably, a modern road still uses this tunnel today. Tunnels were also built for **mineral extraction**. Miners would **locate** a mineral **vein** and then pursue it with shafts and tunnels underground. Traces of such tunnels used to **mine** gold can still be found at the Dolaucothi mines in Wales. When the **sole** purpose of a tunnel was mineral extraction, construction required less planning, as the tunnel route was **determined** by the mineral vein.

Roman tunnel projects were **carefully** planned and **carried out**. The length of time it took to **construct** a tunnel **depended on** the method being used and the type of rock being

remarkably= amazingly, outstandingly, extraordinarily, surprisingly
pass on= convey, send, impart, communicate
agriculture= the practice or science of farming
equip= prepare, train, arm
handhold= a part of something that you can hold onto when climbing it
foothold= a small hole or crack where you can safely put your foot when climbing a steep rock
lid= a cover for the open part of a pot, box, or other container
ensure= guarantee, confirm, certify, warrant, make sure
plumb= exactly upright or level
rod= a long thin pole or bar
determine= decide, conclude, establish, finalize
slope= a piece of ground or a surface that slopes
measure= calculate, compute, quantify, gage
drain= empty, bleed, remove, tap

appear= happen, occur, exist, surface, emerge
end= the part of a place or object that is furthest from its beginning or centre
alternative= other, another, substitute, alternate
advanced= higher, superior, sophisticated, developed
geometry= the study in mathematics of the angles and shapes formed by the relationships of lines, surfaces, and solid objects in space
adjustment= change, alteration, modification, tuning
direction= way, course, track, route, path
encounter= face, meet, run into
deviate= to change what you are doing so that you are not following an expected plan, idea, or type of behaviour
constantly= continuously, frequently, repetitively
penetrate= enter, pierce, infiltrate, breach
usable= practical, serviceable, working, functioning
inscription= writing, caption, engraving
aqueduct= channel, conduit, canal, watercourse
lateral= side, cross, adjacent, sideways
corridor= passage, passageway, hall, hallway
initial= first, early, original
error= mistake, fault, inaccuracy

obstacle= problem, difficulty, hindrance
mineral= a substance that is formed naturally in the earth, such as coal, salt, stone, or gold. Minerals can be dug out of the ground and used
extraction= removal, withdrawal, abstraction, taking out
locate= find, trace, discover, detect
vein= a thin layer of a valuable metal or mineral which is contained in rock
mine= excavate, quarry, dig, extract
sole= only, solitary, single, individual
determine= decide, conclude, establish, finalize

carefully= cautiously, wisely, prudently
carry out= to do something that needs to be organized and planned

excavated. The qanat construction method was usually faster than the counter-excavation method as it was more **straightforward**. This was because the mountain could be excavated not only from the tunnel mouths but also from shafts. The type of rock could also **influence** construction times. When the rock was hard, the Romans **employed** a **technique** called fire **quenching** which **consisted of** heating the rock with fire, and then **suddenly** cooling it with cold water so that it would **crack**. **Progress** through hard rock could be very slow, and it was not **uncommon** for tunnels to take years, if not decades, to be built. Construction marks left on a Roman tunnel in Bologna show that the **rate of advance** through **solid** rock was 30 centimeters per day. In contrast, the rate of advance of the **Claudius** tunnel can be **calculated** at 1.4 meters per day. Most tunnels had inscriptions showing the names of **patrons** who **ordered** construction and sometimes the name of the **architect**. For example, the 1.4-kilometer Cevlik tunnel in Turkey, built to **divert** the floodwater **threatening** the harbor of the **ancient** city of Seleuceia Pieria, had inscriptions on the **entrance**, still **visible** today, that also **indicate** that the tunnel was started in 69 CE and was completed in 81 CE.

construct= build, make, create, erect
depend on= rely on, count on, trust
straightforward= simple and easy to understand ≠ complicated
influence= affect, shape, change, guide
employ= use, utilize
technique= method, system, practice, procedure
quench a fire/flames= to stop a fire from burning
crack= to break or to make something break
consist of= be made of, be made up of, contain, be composed of
suddenly= quickly and unexpectedly
progress= development, growth, advancement improvement
uncommon= rare, unusual, infrequent
rate= speed, tempo, pace
advance= development, improvement, progress
solid= hard or firm, with a fixed shape, and not a liquid or gas
Claudius= (10 BC-54 AD) the emperor of Rome from AD 41 to 54, who made Britain part of the Roman Empire
calculate= compute, analyze, estimate, determine reckon
patron= someone who supports the activities of an organization, for example by giving money
order= request, ask for, command
architect= someone whose job is to design buildings
divert= redirect, deflect, reroute, switch
ancient= early, antique, olden
threaten= to be likely to harm or destroy something
entrance= entry, access, doorway, door ≠ exit
visible= noticeable, observable, perceptible, evident
indicate= specify, show, signpost, direct, point to

TEST 4

READING PASSAGE 2

Changes In Reading Habits

Look around on your next plane trip. The iPad is the new

pacifier for babies and **toddlers**. Younger **school-aged** children read stories on smartphones; older kids don't read at all, but **hunch** over video games. Parents and other **passengers** read on tablets or **skim** a **flotilla** of email and news feeds. **Unbeknown to** most of us, an **invisible, game-changing transformation links** everyone in this picture: the **neuronal circuit** that **underlies** the brain's **ability** to read is **subtly, rapidly** changing and this has **implications** for everyone from the pre-reading toddler to the **expert** adult.

pacifier= a rubber object that you give a baby to suck so that it does not cry
toddler= baby, kid, child
school age= the age at which a child is old enough to go to school
hunch= bend, huddle
passenger= traveler, customer, fare, commuter
flotilla= a group of small ships
skim= read quickly, speed-read, browse
unbeknown to somebody= without that person knowing about it
invisible= unseeable, undetectable, obscure, imperceptible
game-changing= having a big effect on the conditions in an area such as business
transformation= change, alteration, mutation, modification
link= connect, relate, associate
neuronal= relating to a nerve cell or a neuron (= a basic unit of a nerve cell)
circuit= route, course, track, trail, path
underlie= motivate, cause, inspire, trigger
ability= aptitude, skill, capability, capacity
subtly= intelligently, sensitively, artfully
rapidly= fast, quickly, speedily, swiftly
implication= suggestion, inference, association, consequence
expert= having a special skill or special knowledge of a subject ≠ inexpert

As work in **neurosciences indicates**, the **acquisition** of **literacy necessitated** a new circuit in our **species'** brain more than 6,000 years ago. That circuit **evolved** from a very simple **mechanism** for **decoding** basic information, like the number of goats in one's **herd**, to the present, highly elaborated reading brain. My research **depicts** how the present reading brain **enables** the development of some of our most important **intellectual** and **affective processes: internalized** knowledge, analogical **reasoning**, and **inference**; perspective-taking and **empathy**; **critical** analysis and the **generation** of **insight**. Research surfacing in many parts of the world now **cautions** that each of these essential 'deep reading' processes may be under threat as we move into digital-based modes of reading.

This is not a simple, **binary** issue of print versus digital reading and technological **innovation**. As MIT scholar Sherry Turkle has written, we do not **err** as a society when we innovate but when we **ignore** what we **disrupt** or **diminish** while innovating. In this hinge moment between print and **digital** cultures, society needs to **confront** what is diminishing in the expert reading circuit, what our children and older students are not developing, and what we can do about it.

We know from research that the reading circuit is not given to human beings through a **genetic blueprint** like **vision** or language; it needs an environment to develop. Further, it will **adapt** to that environment's **requirements** – from different writing systems to the **characteristics** of whatever **medium** is used. If the **dominant** medium advantages processes that are fast, **multi-task** oriented and well-suited for large **volumes** of information, like the current digital medium, so will the reading circuit. As UCLA psychologist Patricia Greenfield writes, the result is that less attention and time will be **allocated** to slower, time-demanding deep reading processes.

Increasing reports from educators and from researchers in psychology and the **humanities** bear this out. English literature **scholar** and teacher Mark Edmundson describes how many college students actively avoid the classic literature

neuroscience= the scientific study of the brain
indicate= show, suggest, reveal
acquisition= gaining, attainment, achievement, purchase
literacy= the state of being able to read and write
necessitate= require, demand, need, dictate
evolve= change, grow, progress, advance
species= a group of animals or plants whose members are similar and can breed together to produce young animals or plants
mechanism= means, method, system, procedure
decode= make sense of, work out, interpret, translate
herd= group, flock, drove, pack
depict= show, represent, describe, illustrate
enable= allow, permit, aid, empower
intellectual= intelligent, knowledgeable, academic, rational
affective= emotional, sentimental, moving, touching
process= procedure, development, course, progression
internalize= adopt, affect, assume
reasoning= analysis, logic, calculation, thought
inference= implication, interpretation, suggestion
empathy= understanding, sympathy, compassion, responsiveness, identification
critical= analytical, judicious, diagnostic, serious, detailed
generation= production, making, creation, invention
insight= vision, understanding, awareness, perception
caution= warn, alert, notify, signal

binary= consisting of two parts
innovation= finding, discovery, breakthrough
err= to make a mistake
ignore= pay no attention to, take no notice of, overlook, disregard
disrupt= disturb, upset, interrupt
diminish= reduce, lessen, weaken, moderate
digital= numerical, alphanumeric, numerary, numeral
confront= meet, face, encounter, handle

genetic= relating to genes or genetics
blueprint= design, pattern
vision= eyesight, sight, ability to see
adapt= adjust, become accustomed, get used to, familiarize yourself
characteristic= quality, attribute, trait, feature
medium= way, avenue, mode, method, means
dominant= more powerful, important, or noticeable than other people or things
multi-task= to do several things at the same time
requirement= obligation, condition, must, necessity
volume= quantity, amount, degree, size
allocate= assign, allot, apportion, distribute, give, share

humanity= people in general
scholar= an intelligent and well-educated person

of the 19th and 20th centuries in favour of something simpler as they no longer have the **patience** to read longer, **denser**, more difficult texts. We should be less concerned with students' '**cognitive** impatience', however, than by what may underlie it: the **potential** inability of large numbers of students to read with a level of critical analysis **sufficient** to **comprehend** the complexity of thought and argument found in more **demanding** texts.

Multiple studies show that digital screen use may be causing a variety of **troubling downstream** effects on reading comprehension in older high school and college students. In Stavanger, Norway, psychologist Anne Mangen and her **colleagues** studied how high school students comprehend the same material in different mediums. Mangen's group asked subjects questions about a short story whose **plot** had **universal** student **appeal**; half of the students read the story on a tablet, the other half in paperback. Results indicated that students who read on print were **superior** in their comprehension to screen-reading peers, particularly in their ability to **sequence** detail and reconstruct the plot in **chronological** order.

Ziming Liu from San Jose State University has conducted a series of studies which indicate that the 'new **norm**' in reading is skimming, involving word-spotting and **browsing** through the text. Many readers now use a pattern when reading in which they **sample** the first line and then word- **spot** through the rest of the text. When the reading brain skims like this, it reduces time allocated to deep reading processes. In other words, we don't have time to grasp **complexity**, to understand another's feelings, to **perceive** beauty, and to create thoughts of the reader's own.

The **possibility** that critical **analysis**, empathy and other deep reading **processes** could become the **unintended** '**collateral** damage' of our digital culture is not a **straightforward** binary issue about print versus digital reading. It is about how we all have begun to read on **various** mediums and how that changes not only what we read, but also the **purposes** for which we read. Nor is it only about the young. The subtle **atrophy** of critical analysis and empathy affects us all **equally**. It affects our ability to **navigate** a **constant bombardment** of information. It **incentivizes** a retreat to the most familiar stores of unchecked information, which require and receive no analysis, leaving us **susceptible** to false information and **irrational** ideas.

There's an old rule in neuroscience that does not **alter** with age: use it or lose it. It is a very **hopeful principle** when **applied** to critical thought in the reading brain because it

patience= tolerance, persistence, endurance ≠ impatience

dense= crowded, full, thick, jam-packed
cognitive= reasoning, mental, intellectual, perceptive

potential= possible, latent, probable, likely

sufficient= enough, adequate

comprehend= understand, know, grasp

demanding= difficult, hard, challenging, tough, severe

troubling= worrying

downstream= relating to an activity, product etc that depends on or happens after another activity etc

colleague= coworker, partner, teammate, associate

plot= story, storyline, action, outline

universal= worldwide, general, common

appeal= charm, attraction, interest

superior= excellent, high-class, top-quality, exclusive

sequence= order, arrange, structure

chronological= sequential, consecutive, linear

norm= standard, rule, custom

perceive= see, understand, identify, recognize

browse= surf, look, glance

sample= test, try, taste, experiment

spot= see, notice, recognize

complexity= difficulty, intricacy, complication involvedness

possibility= likelihood, opportunity, prospect

analysis= study, investigation, examination, scrutiny

process= procedure, course, development, progression

intend= mean, aim, propose, plan

collateral= relating to something or happening as a result of it, but not as important

various= numerous, many, several, countless

straightforward= simple and easy to understand

purpose= intention, aim, objective

atrophy= weaken, shrivel, degenerate, deteriorate

equally= evenly, equivalently, alike

navigate= direct, steer, circumnavigate

constant= frequent, persistent, recurrent, continual

bombardment= attack, offensive, assault

incentivize= to give someone a reason to do something, especially by offering them a reward

susceptible= prone, disposed, vulnerable, at risk

irrational= illogical, unreasonable, foolish, crazy

alter= change, modify, adjust, vary

hopeful= promising, encouraging, positive

principle= belief, attitude, opinion, value, standard

apply= relate, pertain, affect, concern

redress= equalize, right, rectify, remedy

implies choice. The story of the changing reading brain is hardly finished. We possess both the science and the technology to identify and redress the changes in how we read before they become entrenched. If we work to understand exactly what we will lose, alongside the extraordinary new capacities that the digital world has brought us, there is as much reason for excitement as caution.	imply= suggest, infer, hint at, point toward entrenched= fixed, rooted, engrained, ingrained extraordinary= notable, amazing, outstanding, remarkable possess= have, own, hold, keep identify= classify, recognize, find, detect exactly= accurately, closely, correctly
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TEST 4

READING PASSAGE 3

Attitudes Towards Artificial Intelligence

A Artificial intelligence (AI) can already **predict** the future. Police forces are using it to **map** when and where crime is likely to **occur**. Doctors can use it to predict when a **patient** is most likely to have a heart attack or **stroke**. **Researchers** are even trying to give AI **imagination** so it can plan for **unexpected consequences**. Many decisions in our lives **require** a good forecast, and AI is almost always better at forecasting than we are. Yet for all these technological **advances**, we still seem to deeply **lack confidence** in AI predictions. Recent cases show that people don't like **relying on** AI and prefer to **trust** human **experts**, even if these experts are wrong. If we want AI to really **benefit** people, we need to find a way to get people to trust it. To do that, we need to understand why people are so **reluctant** to trust AI in the first place.

B Take the case of Watson for Oncology, one of technology **giant** IBM's **supercomputer** programs. Their **attempt** to **promote** this program to cancer doctors was a PR **disaster**. The AI promised to **deliver** top-quality **recommendations** on

artificial= false, fake, non-natural, man-made
intelligence= cleverness, aptitude, intellect
predict= forecast, foresee, envisage, expect
map= chart, plot, plan, draw, represent
occur= take place, happen, have effect
patient= someone who is receiving medical treatment from a doctor or in a hospital
stroke= if someone has a stroke, an artery (=tube carrying blood) in their brain suddenly bursts or becomes blocked, so that they may die or be unable to use some muscles
researcher= someone who studies a subject in detail in order to discover new facts or test new ideas
imagination= creativity, originality, inventiveness
unexpected= surprising, unpredicted, unanticipated
consequence= result, effect, outcome
require= demand, expect, necessitate
advance= development, improvement, spread progress
lack= not have, be short of, be deficient in
confidence= sureness, self-assurance, self-reliance
rely on= depend on, count on, trust, be sure of
trust= believe, have faith in, confide in
expert= specialist, professional, authority
benefit= help, promote, profit, aid
reluctant= unwilling, unenthusiastic, disinclined, hesitant

giant= huge, enormous, vast, massive
supercomputer= processor, processor, CPU, mainframe
attempt= effort, try, go
promote= encourage, help, stimulate, support
disaster= tragedy, ruin, adversity, catastrophe
deliver= bring, transport, carry, send
recommendation= advice, proposal, suggestion

the **treatment** of 12 cancers that **accounted for** 80% of the world's cases. But when doctors first **interacted** with Watson, they found themselves in a rather difficult situation. On the one hand, if Watson provided **guidance** about a treatment that **coincided** with their own opinions, physicians did not see much point in Watson's recommendations. The supercomputer was simply telling them what they already knew, and these recommendations did not change the **actual** treatment. On the other hand, if Watson **generated** a recommendation that **contradicted** the experts' opinion, doctors would **typically** conclude that Watson wasn't **competent**. And the machine wouldn't be able to explain why its treatment was **plausible** because its machine-learning **algorithms** were simply too **complex** to be fully understood by humans. **Consequently**, this has caused even more **suspicion** and **disbelief**, leading many doctors to **ignore** the **seemingly outlandish** AI recommendations and **stick** to their own **expertise**.

C This is just one example of people's lack of confidence in AI and their **reluctance** to accept what AI has to offer. Trust in other people is often based on our understanding of how others think and having experience of their **reliability**. This helps create a **psychological** feeling of safety. AI, on the other hand, is still **fairly** new and **unfamiliar** to most people. Even if it can be **technically** explained (and that's not always the case), AI's decision-making **process** is usually too difficult for most people to **comprehend**. And **interacting** with something we don't understand can cause **anxiety** and give us a sense that we're losing **control**. Many people are also simply not familiar with many **instances** of AI actually working, because it often happens in the background. Instead, they are **acutely** aware of instances where AI goes wrong. Embarrassing AI failures receive a **disproportionate** amount of media attention, **emphasising** the message that we cannot rely on technology. Machine learning is not **foolproof**, in part because the humans who design it aren't.

D Feelings about AI run deep. In a recent experiment, people from a range of backgrounds were given **various** sci-fi films about AI to watch and then asked questions about **automation** in everyday life. It was found that, **regardless of** whether the film they watched **depicted** AI in a positive or negative light, simply watching a **cinematic vision** of our technological future **polarised** the participants' **attitudes**. **Optimists** became more

treatment= cure, healing, care, medicine, remedy
account for= comprise, make up, represent, constitute
interact= relate, cooperate, interrelate, work together
guidance= help, assistance, support, direction
coincide= happen together, concur, overlap, agree, match
actual= real, definite, genuine, authentic
generate= make, produce, create, cause
contradict= deny, reverse, oppose, challenge
typically= characteristically, classically, naturally, stereotypically
competent= capable, able, knowledgeable, experienced
plausible= reasonable, believable, credible, probable
algorithm= a set of instructions that are followed in a fixed order and used for solving a mathematical problem, making a computer program etc
complex= difficult, complicated, intricate
consequently= so, thus, therefore, accordingly, as a result
suspicion= doubt, misgiving, thought, distrust
disbelief= doubt, distrust, skepticism, incredulity
ignore= pay no attention to, take no notice of, overlook, disregard
seemingly= apparently, outwardly, ostensibly
outlandish= unusual, bizarre, strange, weird
stick= attach, glue, fix, join
expertise= skill, knowledge, proficiency, knowhow

reluctance= unwillingness, disinclination, hesitancy
reliability= dependability, consistency, steadfastness, trustworthiness
psychological= mental, emotional, spiritual
fairly= quite, moderately, rather, relatively
unfamiliar= new, untried, unknown, strange, alien
technically= precisely, exactly, theoretically, officially
process= procedure, course, development, progression
comprehend= understand, know, grasp
interact= relate, cooperate, interrelate, work together
anxiety= worry, nervousness, concern, unease
control= power, domination, management
instance= example, illustration, representative
acutely= very, intensely, highly, deeply
disproportionate= uneven, unequal, inconsistent, unbalanced
emphasize= stress, highlight, accentuate, underline
foolproof= safe, guaranteed, infallible, perfect

various= numerous, many, several, countless
automation= mechanization, computerization, robotics
regardless of= irrespective of, despite, notwithstanding, no matter
depict= show, represent, describe, illustrate
cinematic= filmic, movielike, photographic
vision= foresight, imagination, prediction
polarize= separate, differentiate, divide, split
attitude= position, stance, manner, viewpoint
optimist= hoper, idealist, romantic, utopian

extreme in their **enthusiasm** for AI and **sceptics** became even more **guarded**. This suggests people use **relevant evidence** about AI in a **biased** manner to support their existing attitudes, a deep-rooted human **tendency** known as “**confirmation bias**”. As AI is **represented** more and more in media and entertainment, it could lead to a society split between those who benefit from AI and those who reject it. More **pertinently**, refusing to accept the advantages offered by AI could place a large group of people at a serious **disadvantage**.

E Fortunately, we already have some ideas about how to **improve** trust in AI. Simply having **previous experience** with AI can **significantly** improve people’s opinions about the technology, as was found in the study mentioned above. Evidence also suggests the more you use other technologies such as the internet, the more you trust them. Another **solution** may be to **reveal** more about the algorithms which AI uses and the **purposes** they **serve**. Several **high-profile** social media companies and online marketplaces already **release transparency** reports about government **requests** and **surveillance disclosures**. A similar practice for AI could help people have a better understanding of the way algorithmic decisions are made.

F Research suggests that allowing people some control over AI decision-making could also improve trust and **enable** AI to learn from human experience. For example, one study showed that when people were allowed the **freedom** to **slightly modify** an algorithm, they felt more **satisfied** with its decisions, more likely to believe it was **superior** and more likely to use it in the future. We don’t need to understand the **intricate inner workings** of AI systems, but if people are given a **degree** of responsibility for how they are **implemented**, they will be more **willing** to accept AI into their lives.

extreme= great, tremendous, severe, intense
enthusiasm= eagerness, interest, passion, keenness
sceptic= doubter, disbeliever, questioner
guarded= cautious, hesitant, careful
relevant= related, applicable, pertinent
evidence= proof, sign, indication
biased= unfair, partial, prejudiced, influenced
tendency= trend, movement, drive, inclination
confirmation= validation, authorization, approval
represent= symbolize, exemplify, denote, embody
pertinently= relevantly, appositely, appropriately, suitably

disadvantage= difficulty, drawback, shortcoming, weakness ≠ advantage

fortunately= luckily, providentially, opportunely
improve= enhance, increase, boost, develop
previous= earlier, prior, former, aforementioned
experience= knowledge, involvement, skill, practice
significantly= considerably, appreciably, drastically, notably
solution= answer, key, resolution
reveal= expose, uncover, show, bare
purpose= intention, aim, objective
serve= function, work, operate, act
high-profile= prominent, prestigious, conspicuous
release= announce, publish, circulate, issue
transparency= openness, clearness, unmistakability, unambiguousness
request= demand, appeal, wish
surveillance= observation, watch, scrutiny
disclosure= revelation, expose, discovery

enable= allow, permit, empower, qualify, aid
freedom= liberty, autonomy, self-determination, choice
slightly= a little, marginally, faintly
modify= change, adapt, adjust, alter
satisfied= pleased, fulfilled, contented, happy
superior= excellent, high-class, top-quality, exclusive
intricate= complicated, complex, sophisticated, tricky
inner= internal, innermost, inside, interior
degree= amount, quantity, level, scale, extent
workings= mechanisms, machineries, works
implement= apply, realize, execute, employ
willing= agreeable, eager, keen, ready